

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\401-T-32001MDS-Rev 01.pvdb

Date of Analysis : Jul 20,2026 10:01am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 168.300 mm OD	
Vessel Design Length, Tangent to Tangent	5290.00 mm
Distance of Bottom Tangent above Grade	4000.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	80 °C
Internal Design Pressure	2.300 MPa
External Design Temperature	80 °C
Maximum Allowable Working Pressure	5.71 MPa
Shop Test Pressure	3.050 MPa
Field Test Pressure	7.418 MPa
Required Minimum Design Metal Temperature	-33.4 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-333 6	...	...	K03006	+IT
Head	SA-333 6	...	...	K03006	+IT
Flange	SA-350 LF2	2	...	K03011	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Sup. Lugs	SA-516 70	...	...	K02700	+IT
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Lug Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom head	2.314	0.00	6.130	3.0000	No	No
Flange-1	2.313	0.00	7.300	3.1750	N/A	No
Flange-2	2.312	0.00	6.380	3.1750	N/A	No
Shell-1	2.311	0.00	5.710	3.0000	N/A	No
Flange -3	2.300	0.00	6.640	3.0000	N/A	No
Flange-4	2.300	0.00	6.570	3.0000	N/A	No
Shell-2	2.300	0.00	5.720	3.0000	N/A	No
Flange-5	2.300	0.00	7.160	3.0000	N/A	No
Flange-6	2.300	0.00	7.200	3.0000	N/A	No
Shell-3	2.300	0.00	5.720	3.0000	N/A	No
Flange-7	2.300	0.00	6.690	3.0000	N/A	No
Flange-8	2.300	0.00	6.110	3.0000	N/A	No

Liquid Level: 1432.63 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.000

Element Types and Properties:



Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	8.2	7.0	4.6	...	1.00	1.00
Body Flg	150.2	97.0	38.0	35.1	33.7	33.7	1.00	1.00
Body Flg	247.2	97.0	38.0	35.1	33.0	33.0	1.00	1.00
Cylinder	1647.2	1400.0	7.0	7.0	4.6	...	1.00	1.00
Body Flg	1747.4	97.0	38.0	35.1	33.0	33.0	1.00	1.00
Body Flg	1844.5	97.0	38.0	35.1	33.3	33.3	1.00	1.00
Cylinder	4440.5	2596.0	7.0	7.0	4.6	...	1.00	1.00
Body Flg	4540.7	97.0	38.0	35.1	33.0	33.0	1.00	1.00
Body Flg	4637.7	97.0	38.0	35.1	33.0	33.0	1.00	1.00
Cylinder	5149.8	512.1	7.0	7.0	4.6	...	1.00	1.00
Body Flg	5250.0	97.0	38.0	35.1	33.0	33.0	1.00	1.00
Body Flg	5290.0	40.0	40.0	40.0	35.1	35.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N4 1"	WRC-107/537	0.860	n/a	Passed
N1 2"	WRC-297	0.992	Shell	Passed
N2 1"	WRC-297	0.992	Shell	Passed

Loads for Foundation/Support Design:

Total Wind Shear on Support	6416.	N
Total Earthquake Shear on Support	2160.	N
Wind Moment on Support	2814327.	N-mm
Earthquake Moment on Support	1151230.	N-mm
Wind Load Case Test Moment on Support	1419227.	N-mm
Earthquake Load Case Test Moment on Support	490499.	N-mm
Test Case Wind Shear on Support	2117.	N
Maximum Initial Bolt Load	5221.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	1082.9	kgm
Shop Test - Fabricated + water (full)	1186.0	kgm
Shipping - Fab. + rem. intls.+ shipping app.	1082.9	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	1230.8	kgm
Empty - Fab. + intls. + details + wgths.	1242.7	kgm
Operating - Empty + operating liquid (no ca)	1273.9	kgm
Field Test - Empty weight + test medium (full)	1330.0	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N4 1"	6.13	...	...	OK	6.12	OK	No Calc[*]
N1 2"	5.71	...	...	OK	6.99	OK	No Calc[*]
K1A 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K1B 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K2A 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K2B 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K3A 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K3B 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K3C 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K3D 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K3E 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K3F 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
K4B 1"	5.71	...	...	OK	6.42	OK	No Calc[*]
N2 1"	5.72	...	...	OK	6.12	OK	No Calc[*]
K4A 1"	5.72	...	...	OK	6.42	OK	No Calc[*]
N3 2"	6.11	...	...	OK	7.52	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5.71 Nozzle : K4B 1"
 Minimum MAWP Shells/Flanges : 5.71 Element : Shell-1
 Minimum MAPnc Shells/Flanges : 6.57 Element : Flange-4

Computed Vessel M.A.W.P. : 5.71 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N4 1"	0.000	0.000	70.800
40	N1 2"	1047.230	180.000	113.600
40	K1A 1"	1350.000	0.000	62.800
40	K1B 1"	400.000	0.000	62.800
40	K2A 1"	1350.000	90.000	62.800
40	K2B 1"	400.000	90.000	62.800
40	K3A 1"	1350.000	180.000	62.800
40	K3B 1"	400.000	180.000	62.800
40	K3C 1"	1350.000	270.000	62.800
40	K3D 1"	400.000	270.000	62.800
40	K3E 1"	1350.000	45.000	62.800
40	K3F 1"	400.000	45.000	62.800
40	K4B 1"	1597.230	180.000	62.800
100	N2 1"	4799.999	0.000	70.800
100	K4A 1"	4799.999	180.000	62.800
120	N3 2"	0.000	0.000	140.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: Flange-8

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N3 2"	56	98.237	103.75	14

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4 1"	1 in	LWN	WNF	41.4	8.000	...	...	215	900
K1A 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K1B 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K2A 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K2B 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K3A 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K3B 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K3C 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K3D 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K3E 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K3F 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
K4B 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
N2 1"	25 mm	Actual	LWN	41.4	8.000	...	...	263	900
K4A 1"	25 mm	Actual	LWN	53.8	14.224	...	...	263	600
N3 2"	50 mm	Actual	WNF	66.0	8.000	...	...	190	900
N1 2"	2 in	LWN	WNF	66.8	8.000	...	...	227	900

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:
 Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4 1"	SA-350 LF2	7.100	10.000	...	...	...
K1A 1"	SA-350 LF2	7.100	10.000	...	...	...
K1B 1"	SA-350 LF2	7.100	10.000	...	...	...
K2A 1"	SA-350 LF2	7.100	10.000	...	...	...
K2B 1"	SA-350 LF2	7.000	10.000	...	...	...
K3A 1"	SA-350 LF2	7.000	10.000	...	...	...
K3B 1"	SA-350 LF2	7.000	10.000	...	...	...
K3C 1"	SA-350 LF2	7.100	10.000	...	...	...
K3D 1"	SA-350 LF2	7.100	10.000	...	...	...
K3E 1"	SA-350 LF2	7.100	10.000	...	...	...
K3F 1"	SA-350 LF2	7.100	10.000	...	...	...
K4B 1"	SA-350 LF2	7.100	10.000	...	...	...
N2 1"	SA-350 LF2	7.100	10.000	...	...	...
K4A 1"	SA-350 LF2	7.100	10.000	...	...	...
N3 2"	SA-350 LF2	40.000	10.000	...	...	...
N1 2"	SA-350 LF2	7.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4 1"	...	0.0	204.79	0.00	Bottom head
K1A 1"	1350.000	0.0	252.00	0.00	Shell-1
K1B 1"	400.000	0.0	252.00	0.00	Shell-1
K2A 1"	1350.000	90.0	252.00	0.00	Shell-1
K2B 1"	400.000	90.0	252.00	0.00	Shell-1



K3A 1"	1350.000	180.0	252.00	0.00	Shell-1
K3B 1"	400.000	180.0	252.00	0.00	Shell-1
K3C 1"	1350.000	270.0	252.00	0.00	Shell-1
K3D 1"	400.000	270.0	252.00	0.00	Shell-1
K3E 1"	1350.000	45.0	252.00	0.00	Shell-1
K3F 1"	400.000	45.0	252.00	0.00	Shell-1
K4B 1"	1597.230	180.0	252.00	0.00	Shell-1
N2 1"	4799.999	0.0	250.00	0.00	Shell-3
K4A 1"	4799.999	180.0	252.00	0.00	Shell-3
N3 2"	...	0.0	150.00	0.00	Flange-8
N1 2"	1047.230	180.0	196.85	0.00	Shell-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 8.2mm THK X 154.3mm ID X 50.0mm	SA-333 6
7	WELD NECK FLANGE: 38.0mm THK X 317.5mm OD	SA-350 LF2
1	CYLINDER: 7.0mm THK X 154.3mm ID X 2596.0mm	SA-333 6
1	CYLINDER: 7.0mm THK X 154.3mm ID X 512.1mm	SA-333 6
1	BLIND FLANGE: 40.0mm THK X 317.5mm OD	SA-350 LF2
1	INSULATION: 88mm X 120mm THK	
7	INSULATION: 97mm X 120mm THK	
1	INSULATION: 1400mm X 120mm THK	
1	PACKING: 2500.000mm	...
1	INSULATION: 2596mm X 120mm THK	
4	GUSSET PLATES: 170.0mm X 86.0mm X 150.0mm X 12.0mm	SA-516 70
2	BOTTOM PLATES: 150.0mm X 170.0mm X 18.0mm	SA-516 70
2	LUG BOLTS: 24.0mm TEMA Metric	SA-193 B7
1	PACKING: 150.000mm	...
1	INSULATION: 512mm X 120mm THK	
1	INSULATION: 38mm X 120mm THK	
4	GASKET: 209.6mm OD X 182.6mm ID	...
48	BODY FLANGE BOLTS: 19.0mm DIA	SA-193 B7
96	NUTS FOR BODY FLANGE BOLTS: 19.0mm DIA	...
2	CYLINDRICAL SEGMENT 7.0mm THK X 192.9mm X 1400.0mm	SA-333 6
1	CYLINDRICAL SEGMENT 7.0mm THK X 99.0mm X 1400.0mm	SA-333 6
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Flange-1	[11]	!	-18	-18	-29	0.771	8.235	1.00	No
Flange-2	[11]	!	-18	-18	-29	0.782	7.000	1.00	No
Flange -3	[11]	!	-18	-18	-29	0.859	7.000	1.00	No
Flange-4	[11]	!	-18	-18	-29	0.869	7.000	1.00	No
Flange-5	[11]	!	-18	-18	-29	0.797	11.000	1.00	No
Flange-6	[11]	!	-18	-18	-29	0.792	11.000	1.00	No
Flange-7	[11]	!	-18	-18	-29	0.775	7.000	1.00	No
Flange-8	[11]	!	-18	-18	-29	0.799	10.000	1.00	No
Bottom head	[10]	!	-46	-46		0.933	7.000	1.00	No
Bottom head	[7]	!	-46	-46		0.764	8.235	1.00	No
Shell-1	[8]	!	-46	-46		1.000	7.000	1.00	No
Shell-2	[8]	!	-46	-46		0.998	7.000	1.00	No
Shell-3	[8]	!	-46	-46		0.998	7.000	1.00	No
N4 1"	[1]	!	-46	-46		0.848	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N1 2"	[1]	!	-46	-46		0.999	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K1A 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K1B 1"	[1]	!	-46	-46		1.000	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K2A 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K2B 1"	[1]	!	-46	-46		1.000	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3A 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3B 1"	[1]	!	-46	-46		1.000	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3C 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3D 1"	[1]	!	-46	-46		1.000	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3E 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3F 1"	[1]	!	-46	-46		1.000	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K4B 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N2 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K4A 1"	[1]	!	-46	-46		0.998	7.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N3 2"	[1]	!	-46	-104		0.237	8.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-104					
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						

Warmest MDMT: -18 -18

Required Minimum Design Metal Temperature -33.4 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.



- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Note: Lug supports should be modeled on an element of their own whose length is just slightly greater than the height of the lug assembly. The vessel does not have to be constructed with this extra element, just modeled this way. This is to insure there is proper mass distribution for these types of models.

Warning:

This is a vertical vessel on support lugs and the earthquake type selected is a building code. Building codes assume all of the mass is located above the point of support. This makes their use on lug supported vessels questionable. It is strongly suggested that based on the seismic zone and applicable parameters, an appropriate lateral (Gx) and (+-) vertical (Gy) acceleration be applied using the G loading option.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.3	MPa
Design Internal Temperature	80.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-33.4	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding

Y

Shop Pressure Test:

Type of Pressure Test User Entered Pressure
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	4000	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117



0.670 0.105
 0.800 0.087
 1.000 0.070
 1.200 0.058
 1.400 0.050
 1.600 0.044
 1.800 0.039
 2.000 0.035
 2.400 0.029
 0.000 0.000

Consider Vertical Acceleration Yes
 M.A.W.P + Static Head to Nozzle Y
 Consider MAP New and Cold in Noz. Design N
 Consider External Loads for Nozzle Des. Y
 Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element# 1/12
 Element From Node 10
 Element To Node 20
 Element Type Elliptical
 Description Bottom head
 Distance "FROM" to "TO" 50 mm
 Element Outside Diameter 168.3 mm
 Element Thickness 7 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 8.2353 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.3 MPa
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.4299999
 Material Name SA-333 6 +IT
 Allowable Stress, Ambient 118 MPa
 Allowable Stress, Operating 118 MPa
 Allowable Stress, Hydrotest 216 MPa
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 224 MPa
 UCS-66 Chart Curve Designation Impact Tested
 External Pressure Chart Name CS-2
 UNS Number K03006
 Product Form Smls. & wld. pipe
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0



Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	-38.575 mm
Height/Length of Liquid	88.575 mm
Liquid Density	0.0009984 kg/cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-38.1 mm
Height/Length of Insulation	88.1 mm
Thickness of Insulation	120 mm
Density	0.00022 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	N4 1"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	25.4 mm
Nozzle Schedule	None
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	78.144 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element#	2/12
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Flange-1
Distance "FROM" to "TO"	97.028 mm
Flange Inside Diameter	154.18 mm
Element Thickness	35.052 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.4299999
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.43 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	2 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	97.028 mm
Liquid Density	0.0009984 kg/cm ³



Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	97.028 mm
Thickness of Insulation	120 mm
Density	0.00022 kg/cm³

Element#	3/12
Element From Node	30
Element To Node	40
Element Type	Flange
Description	Flange-2
Distance "FROM" to "TO"	97.028 mm
Flange Inside Diameter	154.18 mm
Element Thickness	35.052 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.4299999
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	97.028 mm
Liquid Density	0.0009984 kg/cm³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	97.028 mm
Thickness of Insulation	120 mm
Density	0.00022 kg/cm³

Element#	4/12
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	1400 mm
Element Outside Diameter	168.3 mm
Element Thickness	7 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	7 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.4299999
Material Name	SA-333 6 +IT
Allowable Stress, Ambient	118 MPa
Allowable Stress, Operating	118 MPa
Allowable Stress, Hydrotest	216 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	224 MPa



UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. & wld. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1150	mm
Liquid Density	0.0009984	kg/cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1400	mm
Thickness of Insulation	120	mm
Density	0.00022	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1 2"	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.89	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1A 1"	
Dist. from "FROM" Node / Offset dist	1102.8	mm
Nozzle Diameter	25.4	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	54.937	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1B 1"	
Dist. from "FROM" Node / Offset dist	152.77	mm
Nozzle Diameter	25.4	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	54.937	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2A 1"	
Dist. from "FROM" Node / Offset dist	1102.8	mm
Nozzle Diameter	25.4	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K2B 1"
 Dist. from "FROM" Node / Offset dist 152.77 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K3A 1"
 Dist. from "FROM" Node / Offset dist 1102.8 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K3B 1"
 Dist. from "FROM" Node / Offset dist 152.77 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K3C 1"
 Dist. from "FROM" Node / Offset dist 1102.8 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K3D 1"
 Dist. from "FROM" Node / Offset dist 152.77 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT



Element From Node 40
 Detail Type Nozzle
 Detail ID K3E 1"
 Dist. from "FROM" Node / Offset dist 1102.8 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 45.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K3F 1"
 Dist. from "FROM" Node / Offset dist 152.77 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 45.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K4B 1"
 Dist. from "FROM" Node / Offset dist 1350 mm
 Nozzle Diameter 25.4 mm
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 54.937 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element# 5/12
 Element From Node 50
 Element To Node 60
 Element Type Flange
 Description Flange -3
 Distance "FROM" to "TO" 97.028 mm
 Flange Inside Diameter 154.18 mm
 Element Thickness 35.052 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 38 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.3 MPa
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 80 °C
 Effective Diameter Multiplier 1.4299999
 Material Name SA-350 LF2 +IT
 Allowable Stress, Ambient 138 MPa
 Allowable Stress, Operating 138 MPa
 Allowable Stress, Hydrotest 225 MPa
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 230.43 MPa
 UCS-66 Chart Curve Designation Impact Tested
 External Pressure Chart Name CS-2
 UNS Number K03011
 Class / Thickness / Grade 2 ::
 Product Form Forgings
 Perform Flange Stress Calculation (Y/N) Y



Weld is pre-Heated	No
Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	97.028 mm
Thickness of Insulation	120 mm
Density	0.00022 kg/cm ³

Element#	6/12
Element From Node	60
Element To Node	70
Element Type	Flange
Description	Flange-4
Distance "FROM" to "TO"	97.028 mm
Flange Inside Diameter	154.18 mm
Element Thickness	35.052 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.4299999
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	97.028 mm
Thickness of Insulation	120 mm
Density	0.00022 kg/cm ³

Element From Node	60
Detail Type	Weight
Detail ID	Bed sup ring
Dist. from "FROM" Node / Offset dist	77.026 mm
Miscellaneous Weight	19.612 N
Offset from Element Centerline	0 mm

Element#	7/12
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	2596 mm
Element Outside Diameter	168.3 mm
Element Thickness	7 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	7 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.4299999
Material Name	SA-333 6 +IT
Allowable Stress, Ambient	118 MPa
Allowable Stress, Operating	118 MPa
Allowable Stress, Hydrotest	216 MPa
Material Density	0.00775 kg/cm ³



P Number Thickness	32.004	mm
Yield Stress, Operating	224	MPa
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. & wld. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Packing	
Detail ID	Bed	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	2500	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	10.0	
Specific Gravity of Packing Liquid	1.0	
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2596	mm
Thickness of Insulation	120	mm
Density	0.00022	kg/cm ³
Element From Node	70	
Detail Type	Lug	
Detail ID	Support Lugs	
Dist. from "FROM" Node / Offset dist	1180	mm
Number of Lugs	2	
Dist. from OD to Lug Cntrline(dlug)	185	mm
Height of Gusset Plates (hgp)	150	mm
Force Bearing Width (wfb)	204	mm
Weight of Lug	69.906	N
Lug Start Angle (degrees)	0.0	
Element From Node	70	
Detail Type	Weight	
Detail ID	Cleat weight	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	Bed limiter	
Dist. from "FROM" Node / Offset dist	705.54	mm
Miscellaneous Weight	9.806	N
Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	Bed sup grid	
Dist. from "FROM" Node / Offset dist	-0.00189	mm
Miscellaneous Weight	14.709	N
Offset from Element Centerline	0	mm

Element#	8/12	
Element From Node	80	
Element To Node	90	
Element Type	Flange	
Description	Flange-5	
Distance "FROM" to "TO"	97.028	mm
Flange Inside Diameter	154.18	mm
Element Thickness	35.052	mm
Internal Corrosion Allowance	3	mm



Nominal Thickness	38	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.3	MPa
Design Temperature Internal Pressure	80	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.4299999	
Material Name	SA-350 LF2 +IT	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.43	MPa
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03011	
Class / Thickness / Grade	2 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	97.028	mm
Thickness of Insulation	120	mm
Density	0.00022	kg/cm³

Element#	9/12	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	Flange-6	
Distance "FROM" to "TO"	97.028	mm
Flange Inside Diameter	154.18	mm
Element Thickness	35.052	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	38	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.3	MPa
Design Temperature Internal Pressure	80	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.4299999	
Material Name	SA-350 LF2 +IT	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	97.028	mm
Thickness of Insulation	120	mm
Density	0.00022	kg/cm³

Element#	10/12	
Element From Node	100	
Element To Node	110	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	512.1	mm
Element Outside Diameter	168.3	mm
Element Thickness	7	mm



Internal Corrosion Allowance	3	mm
Nominal Thickness	7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.3	MPa
Design Temperature Internal Pressure	80	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.4299999	
Material Name	SA-333 6	+IT
Allowable Stress, Ambient	118	MPa
Allowable Stress, Operating	118	MPa
Allowable Stress, Hydrotest	216	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	224	MPa
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. & wld. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Packing	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	313	mm
Height of Packed Section	150	mm
Density	0.60000E-04	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 100	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	512.1	mm
Thickness of Insulation	120	mm
Density	0.00022	kg/cm³
Element From Node	100	
Detail Type	Nozzle	
Detail ID	N2 1"	
Dist. from "FROM" Node / Offset dist	162.31	mm
Nozzle Diameter	25.4	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	97.583	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	100	
Detail Type	Nozzle	
Detail ID	K4A 1"	
Dist. from "FROM" Node / Offset dist	162.31	mm
Nozzle Diameter	25.4	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	54.937	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	100	
Detail Type	Weight	
Detail ID	N2 Feed pipe	
Dist. from "FROM" Node / Offset dist	162.31	mm



Miscellaneous Weight	4.4127	N
Offset from Element Centerline	0	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Distributor	
Dist. from "FROM" Node / Offset dist	160	mm
Miscellaneous Weight	9.806	N
Offset from Element Centerline	0	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Dist sup	
Dist. from "FROM" Node / Offset dist	150	mm
Miscellaneous Weight	9.806	N
Offset from Element Centerline	0	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Lifting lug 1	
Dist. from "FROM" Node / Offset dist	312.31	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	80	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Lifting lug 2	
Dist. from "FROM" Node / Offset dist	312.31	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	80	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Pipe davit	
Dist. from "FROM" Node / Offset dist	312.31	mm
Miscellaneous Weight	4903	N
Offset from Element Centerline	100	mm

Element#	11/12
Element From Node	110
Element To Node	120
Element Type	Flange
Description	Flange-7
Distance "FROM" to "TO"	97.028 mm
Flange Inside Diameter	154.18 mm
Element Thickness	35.052 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.4299999
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.43 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	2 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



Element From Node	110
Detail Type	Insulation
Detail ID	Ins: 110
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	97.028 mm
Thickness of Insulation	120 mm
Density	0.00022 kg/cm ³

Element#	12/12
Element From Node	120
Element To Node	130
Element Type	Flange
Description	Flange-8
Distance "FROM" to "TO"	40 mm
Flange Outside Diameter	317.5 mm
Element Thickness	40 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	40 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.3 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.4299999
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	120
Detail Type	Insulation
Detail ID	Ins: 120
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	38 mm
Thickness of Insulation	120 mm
Density	0.00022 kg/cm ³

Element From Node	120
Detail Type	Nozzle
Detail ID	N3 2"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	50 mm
Nozzle Schedule	None
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	108.81 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Bottom head	...	...	50	...	50
Flange-1	...	...	150.203	...	97.028
Flange-2	...	...	247.231	...	97.028
Shell-1	...	...	1647.23	...	1400
Flange -3	...	...	1747.43	...	97.028
Flange-4	...	...	1844.46	...	97.028
Shell-2	...	...	4440.46	...	2596
Flange-5	...	...	4540.67	...	97.028
Flange-6	...	...	4637.69	...	97.028
Shell-3	...	...	5149.79	...	512.1
Flange-7	...	...	5250	...	97.028
Flange-8	...	...	5290	...	40

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

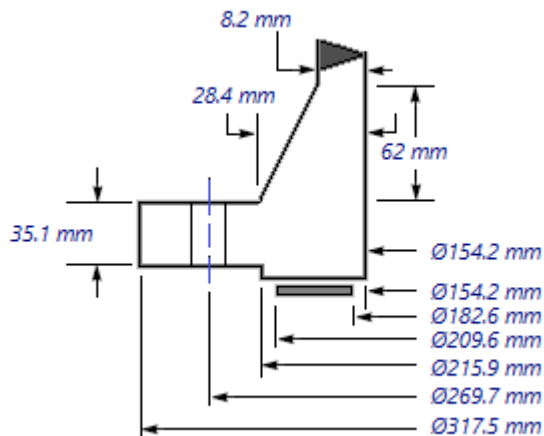
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Flange Input Data Values

Description: FLANGE-1 :

Flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	2.31	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	154.178	mm
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	35.0520	mm
Thickness of Hub at Small End	go	8.2353	mm
Thickness of Hub at Large End	gl	28.4480	mm
Length of Hub	h	61.9760	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	154.178	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		2775.30	N
Bending Moment on Flange		20085.10	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.31 \cdot (154/2 + 3.17)) / (138 \cdot 1 - 0.6 \cdot 2.31) + Ca$$

$$= 4.5341 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 5.06) / (80.3 + 0.6 \cdot 5.06)$$

$$= 8.383 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 8.24) / (77.1 + 0.6 \cdot 8.24)$$

$$= 13.854 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - c_i$	31.877 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	160.528 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	25.273 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	5.060 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	29.337 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	13.462 mm
Basic Gasket Width, $bo = N/2$	6.731 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.538 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 196^2 \cdot 2.31$$

$$= 70125.195 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.31$$

$$= 56002.262 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 161^2 \cdot 2.31$$

$$= 46812.633 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 70125 - 46813$$

$$= 23312.562 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(70125 + 56002 + 0, 0)$$

$$= 126127.453 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((126127 + 2775 + 4 \cdot 20085 / 196) / 172, 278215 / 172)$$

$$= 1617.694 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 31.9 / (3 + 0.5)$$

$$= 92.746 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 270 \cdot \sin(\pi/12)$$

$$= 69.816 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{69.8/(2 \cdot 19 + 31.9)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Hub and Bolts:	28.575	29.337	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	92.746

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi(210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 126127 - 70125$$

$$= 56002.26 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((270 - 161) / 2 + 36.6)$$

$$= 45.6234 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (270 - 161 - 25.3) / 2$$

$$= 41.9735 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 78.5, Bb = 31.9, Cc = 62 \text{ and } Ddg = 15.2 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 78.5 \cdot 31.9^3 \cdot (1/3 - 0.210 \cdot (31.9/78.5) \cdot (1 - 1/12 \cdot (31.9/78.5)^4))$$

$$= 631085.125 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 62 \cdot 15.2^3 \cdot (1/3 - 0.105 \cdot (15.2/62) \cdot (1 - 1/192 \cdot (15.2/62)^4))$$

$$= 66516.938 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 38.5 + 4.06$$

$$= 697602.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 78.5 \cdot 31.9^3 \left(\frac{1}{3} - 0.21 \left(\frac{31.9}{78.5} \right) \left(1 - \frac{1}{12} \left(\frac{31.9}{3.09} \right)^4 \right) \right)$$

$$= 631085.125 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(I_{p,term1}, I_{p,term2})$$

$$= \max(697602, 631085)$$

$$= 697602.1 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.27 \cdot 5.06^2 \cdot 28.5 \cdot 161 / 0.018$$

$$= 707722.500 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	46813.	41.9735	1.0000	1965687.
Face Pressure, Mt	23313.	45.6234	1.0000	1064029.
Gasket Load, Mg	56002.	36.6367	1.0000	2052571.
Gasket Seating, Matm	402105.	36.6367	1.0000	14737776.
Moment Load, Moe	...	...	...	129702.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{I}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 20085 \cdot 707722 / (0.384 \cdot 697602 + 707722) \cdot (42 / (270 - 2 \cdot 42)) + 2775 \cdot 42$$

$$= 129702.234 \text{ N-mm}$$

Total Moment for Operation, Mo 5211989. N-mm
 Total Moment for Gasket seating, Mg 14737776. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 28.501 mm
 Hub Ratio, h/h0 = HL / H0 2.175
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 4.994

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.496
 Factor V per 4.16.5 0.018
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 318 / 161$$

$$= 1.978$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.98^2 \cdot \log_{10}(1.98))) / (1.98^2 - 1) / (1.98 - 1)$$

$$= 3.010$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.98^2(1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / ((1.0472 + 1.9448 \cdot 1.98^2) \cdot (1.98 - 1))$$

$$= 1.515$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.98^2(1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / (1.36136(1.98^2 - 1) \cdot (1.98 - 1))$$

$$= 3.308$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.98^2 + 1) / (1.98^2 - 1)$$

$$= 1.687$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 3.31 \cdot 5.06^2 \cdot 28.5 / 0.018 \\
 &= 131051.891 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.5 / 28.5 \\
 &= 0.017 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3/d \\
 &= (31.9 \cdot 0.017 + 1) / 1.51 + 31.9^3 / 131052 \\
 &= 1.273
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1 \cdot 5211989 / 161) / (1.27 \cdot 25.3^2) \\
 &= 39.91 \text{ MPa}
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1 \cdot 14737776 / 161) / (1.27 \cdot 25.3^2) \\
 &= 112.86 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (1.74 \cdot 5211989 / 161) / (1.27 \cdot 31.9^2) \\
 &= 43.63 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (1.74 \cdot 14737776 / 161) / (1.27 \cdot 31.9^2) \\
 &= 123.37 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (3.01 \cdot 5211989 / (31.9^2 \cdot 161)) - 1.69 \cdot 43.6 \\
 &= 22.54 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (3.01 \cdot 14737776 / (31.9^2 \cdot 161)) - 1.69 \cdot 123 \\
 &= 63.75 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (39.9 + \max(43.6, 22.5)) / 2 \\
 &= 41.77 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (113 + \max(123, 63.7)) / 2 \\
 &= 118.12 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= (Wm1 + Fa + 4Me/G) / Ab \\
 &= 129312 / 2338 \\
 &= 55.31 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (278215 / 2338) \\
 &= 119.00 \text{ MPa}
 \end{aligned}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	39.91	207.00	112.86	207.00
Radial Flange	43.63	138.00	123.37	138.00
Tangential Flange	22.54	138.00	63.75	138.00
Maximum Average	41.77	138.00	118.12	138.00
Bolting	55.31	172.00	119.00	172.00

Minimum Required Flange Thickness 33.731 mm

Estimated M.A.W.P. (Operating) 7.32 MPa

Estimated Finished Weight of Flange at given Thk. 21.2 kgm

Estimated Unfinished Weight of Forging at given Thk 45.5 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14861729 / 1.01 \cdot 1 \cdot 0.018 / (1.23 \cdot 202713 \cdot 5.06^2 \cdot 28.5 \cdot 0.3)$
 $= 0.259$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5255824 / 1.01 \cdot 1 \cdot 0.018 / (1.23 \cdot 199340 \cdot 5.06^2 \cdot 28.5 \cdot 0.3)$
 $= 0.093$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14737776 / 1 \cdot 1 \cdot 0.018 / (1.27 \cdot 202713 \cdot 5.06^2 \cdot 28.5 \cdot 0.3)$
 $= 0.250$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5211989 / 1 \cdot 1 \cdot 0.018 / (1.27 \cdot 199340 \cdot 5.06^2 \cdot 28.5 \cdot 0.3)$
 $= 0.090$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	33.06	207.00	91.20	207.00
Radial Flange	38.30	138.00	105.68	138.00
Tangential Flange	18.30	138.00	50.49	138.00
Maximum Average	35.68	138.00	98.44	138.00
Bolting	55.31	172.00	119.00	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 31.369 mm

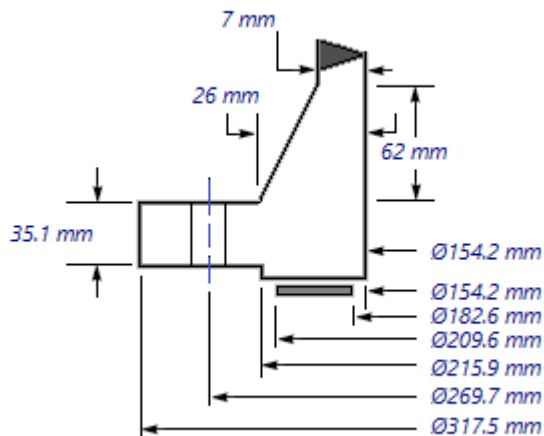
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Flange Input Data Values

Description: FLANGE-2 :

Flange-2

Flange Type	Integral Weld Neck		
Design Pressure	P	2.31	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	154.178	mm
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	35.0520	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	26.0230	mm
Length of Hub	h	61.9760	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	154.178	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		2559.55	N
Bending Moment on Flange		46942.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.31 \cdot (154/2 + 3.17)) / (138 \cdot 1 - 0.6 \cdot 2.31) + Ca$$

$$= 4.5335 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 3.83) / (80.3 + 0.6 \cdot 3.83)$$

$$= 6.393 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 7) / (77.1 + 0.6 \cdot 7)$$

$$= 11.883 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - ci$	31.877 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot ci$	160.528 mm
Corroded Large Hub, $g1_{cor} = g1 - ci$	22.848 mm
Corroded Small Hub, $g0_{cor} = g0 - ci$	3.825 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	31.762 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	13.462 mm
Basic Gasket Width, $bo = N/2$	6.731 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.538 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 196^2 \cdot 2.31$$

$$= 70096.359 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.31$$

$$= 55979.234 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 161^2 \cdot 2.31$$

$$= 46793.383 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 70096 - 46793$$

$$= 23302.980 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(70096 + 55979 + 0, 0)$$

$$= 126075.594 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((126076 + 2560 + 4 \cdot 46942/196) / 172, 278215/172)$$

$$= 1617.694 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 31.9/(3 + 0.5)$$

$$= 92.746 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 270 \cdot \sin(\pi/12)$$

$$= 69.816 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{69.8/(2 \cdot 19 + 31.9)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Hub and Bolts:	28.575	31.762	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	92.746

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi(210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 126076 - 70096$$

$$= 55979.23 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((270 - 161) / 2 + 36.6)$$

$$= 45.6234 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (270 - 161 - 22.8) / 2$$

$$= 43.1860 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 78.5, Bb = 31.9, Cc = 62 \text{ and } Ddg = 13.3 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 78.5 \cdot 31.9^3 \cdot (1/3 - 0.210 \cdot (31.9/78.5) \cdot (1 - 1/12 \cdot (31.9/78.5)^4))$$

$$= 631085.125 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 62 \cdot 13.3^3 \cdot (1/3 - 0.105 \cdot (13.3/62) \cdot (1 - 1/192 \cdot (13.3/62)^4))$$

$$= 45681.906 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 38.5 + 2.79$$

$$= 676767.062 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:



$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 78.5 \cdot 31.9^3 (1/3 - 0.21 (31.9/78.5) (1 - 1/12 (31.9/3.09)^4))$$

$$= 631085.125 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(I_p, \text{term1}, I_p, \text{term2})$$

$$= \max(676767, 631085)$$

$$= 676767.1 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.42 \cdot 3.83^2 \cdot 24.8 \cdot 161 / 0.013$$

$$= 544985.438 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	46793.	43.1860	1.0000	2021638.
Face Pressure, Mt	23303.	45.6234	1.0000	1063591.
Gasket Load, Mg	55979.	36.6367	1.0000	2051727.
Gasket Seating, Matm	402105.	36.6367	1.0000	14737776.
Moment Load, Moe	...	...	...	140524.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 46942 [544985 / (0.384 \cdot 676767 + 544985)] \cdot (43.2 / (270 - 2 \cdot 43.2)) + 2560 \cdot 43.2$$

$$= 140523.797 \text{ N-mm}$$

Total Moment for Operation, Mo 5277480. N-mm
 Total Moment for Gasket seating, Mg 14737776. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 24.779 mm
 Hub Ratio, h/h0 = HL / H0 2.501
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 5.973

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.474
 Factor V per 4.16.5 0.013
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 318 / 161$$

$$= 1.978$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.98^2 \cdot \log_{10}(1.98))) / (1.98^2 - 1)) / (1.98 - 1)$$

$$= 3.010$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / ((1.0472 + 1.9448 \cdot 1.98^2) \cdot (1.98 - 1))$$

$$= 1.515$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / (1.36136 (1.98^2 - 1) \cdot (1.98 - 1))$$

$$= 3.308$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.98^2 + 1) / (1.98^2 - 1)$$

$$= 1.687$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.31 \cdot 3.83^2 \cdot 24.8 / 0.013$$

$$= 90456.930 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.47 / 24.8$$

$$= 0.019 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (31.9 \cdot 0.019 + 1) / 1.51 + 31.9^3/90457$$

$$= 1.420$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 5277480 / 161) / (1.42 \cdot 22.8^2)$$

$$= 44.32 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 14737776 / 161) / (1.42 \cdot 22.8^2)$$

$$= 123.78 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 5277480 / 161) / (1.42 \cdot 31.9^2)$$

$$= 41.27 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 14737776 / 161) / (1.42 \cdot 31.9^2)$$

$$= 115.25 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.01 \cdot 5277480 / (31.9^2 \cdot 161)) - 1.69 \cdot 41.3$$

$$= 27.74 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.01 \cdot 14737776 / (31.9^2 \cdot 161)) - 1.69 \cdot 115$$

$$= 77.46 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (44.3 + \max(41.3, 27.7)) / 2$$

$$= 42.80 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (124 + \max(115, 77.5)) / 2$$

$$= 119.51 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm_1 + Fa + 4Me/G) / Ab$$

$$= 129590 / 2338$$

$$= 55.43 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm_2 / Ab)$$

$$= (278215 / 2338)$$

$$= 119.00 \text{ MPa}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	44.32	207.00	123.78	207.00
Radial Flange	41.27	138.00	115.25	138.00
Tangential Flange	27.74	138.00	77.46	138.00
Maximum Average	42.80	138.00	119.51	138.00
Bolting	55.43	172.00	119.00	172.00

Minimum Required Flange Thickness 33.045 mm

Estimated M.A.W.P. (Operating) 6.39 MPa

Estimated Finished Weight of Flange at given Thk. 20.7 kgm

Estimated Unfinished Weight of Forging at given Thk 45.5 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14936513 / 1.01 \cdot 1 \cdot 0.013 / (1.33 \cdot 202713 \cdot 3.83^2 \cdot 24.8 \cdot 0.3)$
 $= 0.347$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5348647 / 1.01 \cdot 1 \cdot 0.013 / (1.33 \cdot 199340 \cdot 3.83^2 \cdot 24.8 \cdot 0.3)$
 $= 0.126$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14737776 / 1 \cdot 1 \cdot 0.013 / (1.42 \cdot 202713 \cdot 3.83^2 \cdot 24.8 \cdot 0.3)$
 $= 0.325$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5277480 / 1 \cdot 1 \cdot 0.013 / (1.42 \cdot 199340 \cdot 3.83^2 \cdot 24.8 \cdot 0.3)$
 $= 0.118$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	37.98	207.00	103.57	207.00
Radial Flange	37.36	138.00	101.87	138.00
Tangential Flange	20.78	138.00	56.65	138.00
Maximum Average	37.67	138.00	102.72	138.00
Bolting	55.43	172.00	119.00	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 31.013 mm

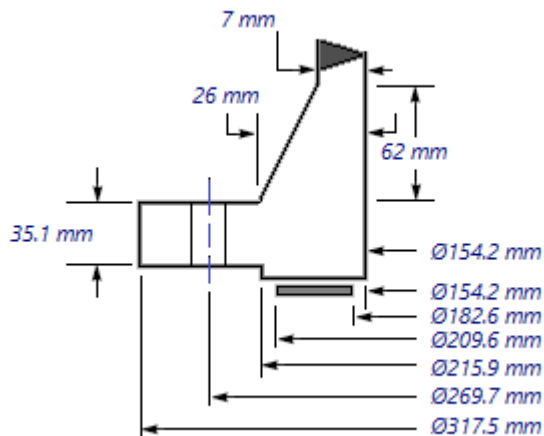
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Flange Input Data Values

Description: FLANGE-3 :

Flange -3

Flange Type	Integral Weld Neck		
Design Pressure	P	2.30	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	154.178	mm
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	35.0520	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	26.0230	mm
Length of Hub	h	61.9760	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	154.178	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		1961.34	N
Bending Moment on Flange		1860900.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.3 \cdot (154/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 2.3) + Ca$$

$$= 4.3483 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 4) / (80.1 + 0.6 \cdot 4)$$

$$= 6.692 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 7) / (77.1 + 0.6 \cdot 7)$$

$$= 11.883 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - ci$	32.052 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot ci$	160.178 mm
Corroded Large Hub, $g1_{cor} = g1 - ci$	23.023 mm
Corroded Small Hub, $g0_{cor} = go - ci$	4.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	31.762 mm
Gasket Contact Width, $N = (Go - Gi)/2$	13.462 mm
Basic Gasket Width, $bo = N/2$	6.731 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.538 mm
Gasket Reaction Diameter, $G = Go - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 196^2 \cdot 2.3$$

$$= 69725.781 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.3$$

$$= 55683.285 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 160^2 \cdot 2.3$$

$$= 46343.246 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 69726 - 46343$$

$$= 23382.537 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(69726 + 55683 + 0, 0)$$

$$= 125409.062 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((125409 + 1961 + 4 \cdot 1860900/196) / 172, 278215/172)$$

$$= 1617.694 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 32.1/(3 + 0.5)$$

$$= 93.046 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 270 \cdot \sin(\pi/12)$$

$$= 69.816 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{69.8/(2 \cdot 19 + 32.1)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Hub and Bolts:	28.575	31.762	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	93.046

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi(210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 125409 - 69726$$

$$= 55683.28 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((270 - 160) / 2 + 36.6)$$

$$= 45.7109 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (270 - 160 - 23) / 2$$

$$= 43.2735 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 78.7, Bb = 32.1, Cc = 62 \text{ and } Ddg = 13.5 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 78.7 \cdot 32.1^3 \cdot (1/3 - 0.210 \cdot (32.1/78.7) \cdot (1 - 1/12 \cdot (32.1/78.7)^4))$$

$$= 642256.062 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 62 \cdot 13.5^3 \cdot (1/3 - 0.105 \cdot (13.5/62) \cdot (1 - 1/192 \cdot (13.5/62)^4))$$

$$= 47458.582 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 39.2 + 2.9$$

$$= 689714.625 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 78.7 \cdot 32.1^3 (1/3 - 0.21 (32.1/78.7) (1 - 1/12 (32.1/3.1)^4))$$

$$= 642256.062 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(I_p, \text{term1}, I_p, \text{term2})$$

$$= \max(689715, 642256)$$

$$= 689714.6 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.41 \cdot 4^2 \cdot 25.3 \cdot 160 / 0.014$$

$$= 564323.500 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	46343.	43.2735	1.0000	2006248.
Face Pressure, Mt	23383.	45.7109	1.0000	1069269.
Gasket Load, Mg	55683.	36.6367	1.0000	2040880.
Gasket Seating, Matm	402105.	36.6367	1.0000	14737776.
Moment Load, Moe	...	...	...	1281538.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 1860900 [564324 / (0.384 \cdot 689715 + 564324)] \cdot (43.3 / (270 - 2 \cdot 43.3)) + 1961 \cdot 43.3$$

$$= 1281538.250 \text{ N-mm}$$

Total Moment for Operation, Mo 6397936. N-mm
 Total Moment for Gasket seating, Mg 14737776. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 25.312 mm
 Hub Ratio, h/h0 = HL / H0 2.448
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 5.756

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.478
 Factor V per 4.16.5 0.014
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 318 / 160$$

$$= 1.982$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.98^2 \cdot \log_{10}(1.98))) / (1.98^2 - 1)) / (1.98 - 1)$$

$$= 3.001$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / ((1.0472 + 1.9448 \cdot 1.98^2) \cdot (1.98 - 1))$$

$$= 1.513$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / (1.36136 (1.98^2 - 1) \cdot (1.98 - 1))$$

$$= 3.297$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.98^2 + 1) / (1.98^2 - 1)$$

$$= 1.683$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 3.3 \cdot 4^2 \cdot 25.3 / 0.014 \\
 &= 94269.492 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.48 / 25.3 \\
 &= 0.019 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3/d \\
 &= (32.1 \cdot 0.019 + 1) / 1.51 + 32.1^3 / 94269 \\
 &= 1.410
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1 \cdot 6397935 / 160) / (1.41 \cdot 23^2) \\
 &= 53.43 \text{ MPa}
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1 \cdot 14737776 / 160) / (1.41 \cdot 23^2) \\
 &= 123.07 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (1.81 \cdot 6397935 / 160) / (1.41 \cdot 32.1^2) \\
 &= 49.81 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (1.81 \cdot 14737776 / 160) / (1.41 \cdot 32.1^2) \\
 &= 114.73 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (3 \cdot 6397935 / (32.1^2 \cdot 160)) - 1.68 \cdot 49.8 \\
 &= 32.81 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (3 \cdot 14737776 / (32.1^2 \cdot 160)) - 1.68 \cdot 115 \\
 &= 75.58 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (53.4 + \max(49.8, 32.8)) / 2 \\
 &= 51.62 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (123 + \max(115, 75.6)) / 2 \\
 &= 118.90 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= (Wm1 + Fa + 4Me/G) / Ab \\
 &= 165241 / 2338 \\
 &= 70.68 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (278215 / 2338) \\
 &= 119.00 \text{ MPa}
 \end{aligned}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	53.43	207.00	123.07	207.00
Radial Flange	49.81	138.00	114.73	138.00
Tangential Flange	32.81	138.00	75.58	138.00
Maximum Average	51.62	138.00	118.90	138.00
Bolting	70.68	172.00	119.00	172.00

Minimum Required Flange Thickness 32.969 mm

Estimated M.A.W.P. (Operating) 6.64 MPa

Estimated Finished Weight of Flange at given Thk. 20.7 kgm

Estimated Unfinished Weight of Forging at given Thk 45.5 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14925671 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.334$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 6479504 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.147$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14737776 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.313$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 6397935 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.138$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	46.53	207.00	103.57	207.00
Radial Flange	45.77	138.00	101.87	138.00
Tangential Flange	25.45	138.00	56.65	138.00
Maximum Average	46.15	138.00	102.72	138.00
Bolting	70.68	172.00	119.00	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 31.013 mm

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Warnings for Flange : FLANGE-4 :

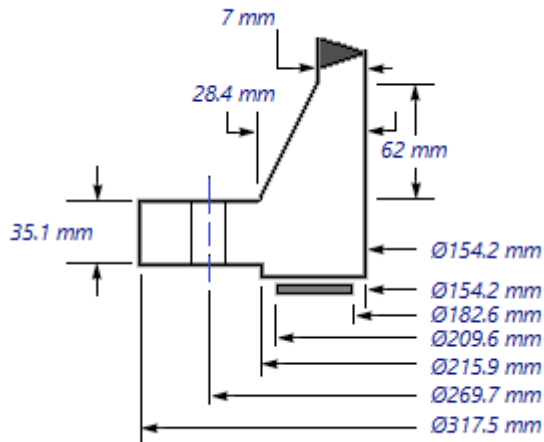
Note: Flange Hub Taper 0.3461 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-4 :

Flange-4

Flange Type	Integral Weld Neck		
Design Pressure	P	2.30	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	154.178	mm
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	35.0520	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	28.4480	mm
Length of Hub	h	61.9760	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	154.178	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		1745.59	N
Bending Moment on Flange		2076900.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.3 \cdot (154/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 2.3) + Ca$
 $= 4.3483 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 4) / (80.1 + 0.6 \cdot 4)$
 $= 6.692 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 7) / (77.1 + 0.6 \cdot 7)$
 $= 11.883 \text{ MPa}$

Corroded Flange Thickness, $t_c = T - c_i$	32.052 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	160.178 mm
Corroded Large Hub, $g_{lCor} = g_l - c_i$	25.448 mm
Corroded Small Hub, $g_{oCor} = g_o - c_i$	4.000 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g_{lCor}$	29.337 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	13.462 mm
Basic Gasket Width, $b_o = N / 2$	6.731 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.538 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi / 4 \cdot G^2 \cdot P$
 $= \pi / 4 \cdot 196^2 \cdot 2.3$
 $= 69725.781 \text{ N}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.3$
 $= 55683.285 \text{ N}$

Hydrostatic End Load at Flange ID [Hd]:
 $= \pi / 4 \cdot B_{cor}^2 \cdot P$
 $= \pi / 4 \cdot 160^2 \cdot 2.3$
 $= 46343.246 \text{ N}$

Pressure Force on Flange Face [Ht]:
 $= H - H_d$
 $= 69726 - 46343$
 $= 23382.537 \text{ N}$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(69726 + 55683 + 0, 0)$$

$$= 125409.062 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((125409 + 1746 + 4 \cdot 2076900 / 196) / 172, 278215 / 172)$$

$$= 1617.694 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 32.1 / (3 + 0.5)$$

$$= 93.046 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 270 \cdot \sin(\pi / 12)$$

$$= 69.816 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 69.8 / (2 \cdot 19 + 32.1) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Hub and Bolts:	28.575	29.337	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	93.046

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi (210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 125409 - 69726$$

$$= 55683.28 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((270 - 160) / 2 + 36.6)$$

$$= 45.7109 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (270 - 160 - 25.4) / 2$$

= 42.0610 mm

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

Aa = 78.7, Bb = 32.1, Cc = 62 and Ddg = 14.7 mm

Compute the value of [Kab]:

= $Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$
 = $78.7 \cdot 32.1^3 \cdot (1/3 - 0.210 \cdot (32.1/78.7) \cdot (1 - 1/12 \cdot (32.1/78.7)^4))$
 = 642256.062 mm⁴

Compute the value of [Kcd]:

= $Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$
 = $62 \cdot 14.7^3 \cdot (1/3 - 0.105 \cdot (14.7/62) \cdot (1 - 1/192 \cdot (14.7/62)^4))$
 = 61009.633 mm⁴

Flange Inertia Term [Ip,term1]:

= Kab + Kcd = 39.2 + 3.72
 = 703265.688 mm⁴

Flange Inertia Term [Ip,term2]:

= $Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$
 = $78.7 \cdot 32.1^3 \cdot (1/3 - 0.21 \cdot (32.1/78.7) \cdot (1 - 1/12 \cdot (32.1/78.7)^4))$
 = 642256.062 mm⁴

Flange Inertia [Ip]:

= max(Ip,term1, Ip,term2)
 = max(703266, 642256)
 = 703265.7 mm⁴

Flange Moment of Inertia for Integral Types [I]:

= $0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$
 = $0.0874 \cdot 1.34 \cdot 4^2 \cdot 25.3 \cdot 160 / 0.012$
 = 641816.812 mm⁴

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	46343.	42.0610	1.0000	1950034.
Face Pressure, Mt	23383.	45.7109	1.0000	1069269.
Gasket Load, Mg	55683.	36.6367	1.0000	2040880.
Gasket Seating, Matm	402105.	36.6367	1.0000	14737776.
Moment Load, Moe	...	...	...	1398385.

Moment Value, eq. 4.16.16, [Moe]:

= $4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 = $4 \cdot 2076900 [641817 / (0.384 \cdot 703266 + 641817)] \cdot (42.1 / (270 - 2 \cdot 42.1)) + 1746 \cdot 42.1$
 = 1398384.875 N-mm

Total Moment for Operation, Mo 6458568. N-mm
 Total Moment for Gasket seating, Mg 14737776. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 25.312 mm
 Hub Ratio, h/h0 = HL / H0 2.448
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 6.362

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.467
 Factor V per 4.16.5 0.012
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

= A / B

$$= 318/160$$

$$= 1.982$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.98^2 \cdot \log_{10}(1.98))) / (1.98^2 - 1)) / (1.98 - 1)$$

$$= 3.001$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.98^2(1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / ((1.0472 + 1.9448 \cdot 1.98^2) \cdot (1.98 - 1))$$

$$= 1.513$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.98^2(1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / (1.36136(1.98^2 - 1) \cdot (1.98 - 1))$$

$$= 3.297$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.98^2 + 1) / (1.98^2 - 1)$$

$$= 1.683$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.3 \cdot 4^2 \cdot 25.3 / 0.012$$

$$= 112480.820 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.47 / 25.3$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (32.1 \cdot 0.018 + 1) / (1.51 + 32.1^3 / 112481)$$

$$= 1.344$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 6458568 / 160) / (1.34 \cdot 25.4^2)$$

$$= 46.31 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 14737776 / 160) / (1.34 \cdot 25.4^2)$$

$$= 105.68 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.79 \cdot 6458568 / 160) / (1.34 \cdot 32.1^2)$$

$$= 52.19 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.79 \cdot 14737776 / 160) / (1.34 \cdot 32.1^2)$$

$$= 119.10 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3 \cdot 6458568 / (32.1^2 \cdot 160)) - 1.68 \cdot 52.2$$

$$= 29.90 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3 \cdot 14737776 / (32.1^2 \cdot 160)) - 1.68 \cdot 119$$

$$= 68.23 \text{ MPa}$$



Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (46.3 + \max(52.2, 29.9)) / 2$$

$$= 49.25 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (106 + \max(119, 68.2)) / 2$$

$$= 112.39 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (W_{m1} + F_a + 4M_e/G) / A_b$$

$$= 169421/2338$$

$$= 72.47 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (278215/2338)$$

$$= 119.00 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	46.31	207.00	105.68 207.00
Radial Flange	52.19	138.00	119.10 138.00
Tangential Flange	29.90	138.00	68.23 138.00
Maximum Average	49.25	138.00	112.39 138.00
Bolting	72.47	172.00	119.00 172.00

Minimum Required Flange Thickness 33.325 mm

Estimated M.A.W.P. (Operating) 6.57 MPa

Estimated Finished Weight of Flange at given Thk. 21.0 kgm

Estimated Unfinished Weight of Forging at given Thk 45.5 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 14886837 / 1.01 \cdot 1 \cdot 0.012 / (1.28 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$$

$$= 0.289 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 6523891 / 1.01 \cdot 1 \cdot 0.012 / (1.28 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$$

$$= 0.129 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 14737776 / 1 \cdot 1 \cdot 0.012 / (1.34 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$$

$$= 0.275 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 6458568 / 1 \cdot 1 \cdot 0.012 / (1.34 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$$

$$= 0.122 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	41.07	207.00	90.42 207.00
Radial Flange	47.76	138.00	105.13 138.00
Tangential Flange	23.35	138.00	51.39 138.00
Maximum Average	44.41	138.00	97.77 138.00
Bolting	72.47	172.00	119.00 172.00

Minimum Req'd Thickness, ambient, no Corrosion: 31.318 mm

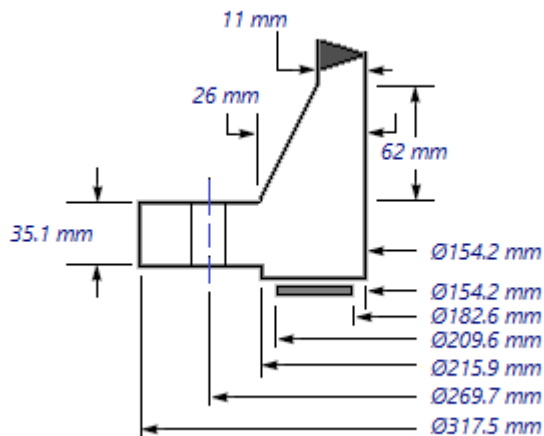
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Flange Input Data Values

Description: FLANGE-5 :

Flange-5

Flange Type	Integral Weld Neck		
Design Pressure	P	2.30	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	154.178	mm
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	35.0520	mm
Thickness of Hub at Small End	go	11.0000	mm
Thickness of Hub at Large End	gl	26.0230	mm
Length of Hub	h	61.9760	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	154.178	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		823.76	N
Bending Moment on Flange		542175.00	N-mm



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.3 \cdot (154/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 2.3) + Ca$$

$$= 4.3483 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 8) / (80.1 + 0.6 \cdot 8)$$

$$= 13.005 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (77.1 + 0.6 \cdot 11)$$

$$= 18.138 \text{ MPa}$$

The hub small end thickness is now taken as the attached shell thickness.

Corroded Flange Thickness, $t_c = T - c_i$	32.052 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	160.178 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	23.023 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	4.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	31.762 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	13.462 mm
Basic Gasket Width, $bo = N/2$	6.731 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.538 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 196^2 \cdot 2.3$$

$$= 69725.781 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.3$$

$$= 55683.285 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 160^2 \cdot 2.3$$

$$= 46343.246 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 69726 - 46343$$

$$= 23382.537 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(69726 + 55683 + 0, 0)$$

$$= 125409.062 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((125409 + 824 + 4 \cdot 542175 / 196) / 172, 278215 / 172)$$

$$= 1617.694 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 32.1 / (3 + 0.5)$$

$$= 93.046 \text{ mm}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 270 \cdot \sin(\pi / 12)$$

$$= 69.816 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{69.8 / (2 \cdot 19 + 32.1)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Hub and Bolts:	28.575	31.762	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	93.046

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi(210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 125409 - 69726$$

$$= 55683.28 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((270 - 160) / 2 + 36.6)$$

$$= 45.7109 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (270 - 160 - 23) / 2$$

$$= 43.2735 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 78.7, Bb = 32.1, Cc = 62 \text{ and } Ddg = 13.5 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 78.7 \cdot 32.1^3 \cdot (1/3 - 0.210 \cdot (32.1/78.7) \cdot (1 - 1/12 \cdot (32.1/78.7)^4))$$

$$= 642256.062 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 62 \cdot 13.5^3 \cdot (1/3 - 0.105 \cdot (13.5/62) \cdot (1 - 1/192 \cdot (13.5/62)^4))$$

$$= 47458.582 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 39.2 + 2.9$$

$$= 689714.625 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:
 $= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$
 $= 78.7 \cdot 32.1^3 (1/3 - 0.21 (32.1/78.7) (1 - 1/12 (32.1/3.1)^4))$
 $= 642256.062 \text{ mm}^4$

Flange Inertia [Ip]:
 $= \max(Ip,term1, Ip,term2)$
 $= \max(689715, 642256)$
 $= 689714.6 \text{ mm}^4$

Flange Moment of Inertia for Integral Types [I]:
 $= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$
 $= 0.0874 \cdot 1.41 \cdot 4^2 \cdot 25.3 \cdot 160 / 0.014$
 $= 564323.500 \text{ mm}^4$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	46343.	43.2735	1.0000	2006248.
Face Pressure, Mt	23383.	45.7109	1.0000	1069269.
Gasket Load, Mg	55683.	36.6367	1.0000	2040880.
Gasket Seating, Matm	402105.	36.6367	1.0000	14737776.
Moment Load, Moe	...	...	...	384301.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 542175 [564324 / (0.384 \cdot 689715 + 564324)] \cdot (43.3 / (270 - 2 \cdot 43.3)) +$
 $824 \cdot 43.3$
 $= 384300.812 \text{ N-mm}$

Total Moment for Operation, Mo 5500698. N-mm
 Total Moment for Gasket seating, Mg 14737776. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 25.312 mm
 Hub Ratio, h/h0 = HL / H0 2.448
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 5.756

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.478
 Factor V per 4.16.5 0.014
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:
 $= A / B$
 $= 318/160$
 $= 1.982$

Stress factor from Table 4.16.4 [Y]:
 $= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.98^2 \cdot \log_{10}(1.98))) / (1.98^2 - 1)) / (1.98 - 1)$
 $= 3.001$

Stress factor from Table 4.16.4 [T]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / ((1.0472 + 1.9448 \cdot 1.98^2) \cdot (1.98 - 1))$
 $= 1.513$

Stress factor from Table 4.16.4 [U]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / (1.36136 (1.98^2 - 1) \cdot (1.98 - 1))$
 $= 3.297$

Stress factor from Table 4.16.4 [Z]:
 $= (K^2 + 1) / (K^2 - 1)$

$$= (1.98^2 + 1) / (1.98^2 - 1)$$

$$= 1.683$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.3 \cdot 4^2 \cdot 25.3 / 0.014$$

$$= 94269.492 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.48 / 25.3$$

$$= 0.019 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (32.1 \cdot 0.019 + 1) / (1.51 + 32.1^3 / 94269)$$

$$= 1.410$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 5500698 / 160) / (1.41 \cdot 23^2)$$

$$= 45.93 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 14737776 / 160) / (1.41 \cdot 23^2)$$

$$= 123.07 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (1.81 \cdot 5500698 / 160) / (1.41 \cdot 32.1^2)$$

$$= 42.82 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (1.81 \cdot 14737776 / 160) / (1.41 \cdot 32.1^2)$$

$$= 114.73 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (3 \cdot 5500698 / (32.1^2 \cdot 160)) - 1.68 \cdot 42.8$$

$$= 28.21 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (Y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (3 \cdot 14737776 / (32.1^2 \cdot 160)) - 1.68 \cdot 115$$

$$= 75.58 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (45.9 + \max(42.8, 28.2)) / 2$$

$$= 44.38 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (123 + \max(115, 75.6)) / 2$$

$$= 118.90 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (W_{m1} + F_a + 4M_e/G) / A_b$$

$$= 137266 / 2338$$

$$= 58.71 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (278215 / 2338)$$

$$= 119.00 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	45.93	207.00	123.07	207.00
Radial Flange	42.82	138.00	114.73	138.00
Tangential Flange	28.21	138.00	75.58	138.00
Maximum Average	44.38	138.00	118.90	138.00
Bolting	58.71	172.00	119.00	172.00

Minimum Required Flange Thickness 32.969 mm

Estimated M.A.W.P. (Operating) 7.16 MPa
 Estimated Finished Weight of Flange at given Thk. 20.7 kgm
 Estimated Unfinished Weight of Forging at given Thk 45.5 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14925671 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.334$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5570828 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.127$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14737776 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.313$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5500698 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.118$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	39.68	207.00	103.57	207.00
Radial Flange	39.03	138.00	101.87	138.00
Tangential Flange	21.70	138.00	56.65	138.00
Maximum Average	39.35	138.00	102.72	138.00
Bolting	58.71	172.00	119.00	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 31.013 mm

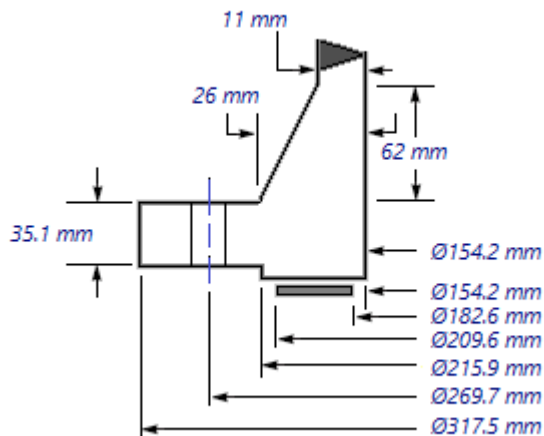
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Flange Input Data Values

Description: FLANGE-6 :

Flange-6

Flange Type	Integral Weld Neck		
Design Pressure	P	2.30	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	154.178	mm
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	35.0520	mm
Thickness of Hub at Small End	go	11.0000	mm
Thickness of Hub at Large End	gl	26.0230	mm
Length of Hub	h	61.9760	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	154.178	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		598.21	N
Bending Moment on Flange		429150.00	N-mm



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.3 \cdot (154/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 2.3) + Ca$$

$$= 4.3483 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 8) / (80.1 + 0.6 \cdot 8)$$

$$= 13.005 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (77.1 + 0.6 \cdot 11)$$

$$= 18.138 \text{ MPa}$$

The hub small end thickness is now taken as the attached shell thickness.

Corroded Flange Thickness, $t_c = T - c_i$	32.052 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	160.178 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	23.023 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	4.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	31.762 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	13.462 mm
Basic Gasket Width, $bo = N/2$	6.731 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.538 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 196^2 \cdot 2.3$$

$$= 69725.781 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.3$$

$$= 55683.285 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 160^2 \cdot 2.3$$

$$= 46343.246 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 69726 - 46343$$

$$= 23382.537 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(69726 + 55683 + 0, 0)$$

$$= 125409.062 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((125409 + 598 + 4 \cdot 429150/196)/172, 278215/172)$$

$$= 1617.694 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 32.1/(3 + 0.5)$$

$$= 93.046 \text{ mm}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 270 \cdot \sin(\pi / 12)$$

$$= 69.816 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{69.8 / (2 \cdot 19 + 32.1)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Hub and Bolts:	28.575	31.762	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	93.046

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi(210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 125409 - 69726$$

$$= 55683.28 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((270 - 160) / 2 + 36.6)$$

$$= 45.7109 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (270 - 160 - 23) / 2$$

$$= 43.2735 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 78.7, Bb = 32.1, Cc = 62 \text{ and } Ddg = 13.5 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 78.7 \cdot 32.1^3 \cdot (1/3 - 0.210 \cdot (32.1/78.7) \cdot (1 - 1/12 \cdot (32.1/78.7)^4))$$

$$= 642256.062 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 62 \cdot 13.5^3 \cdot (1/3 - 0.105 \cdot (13.5/62) \cdot (1 - 1/192 \cdot (13.5/62)^4))$$

$$= 47458.582 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 39.2 + 2.9$$

$$= 689714.625 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:
 $= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$
 $= 78.7 \cdot 32.1^3 (1/3 - 0.21 (32.1/78.7) (1 - 1/12 (32.1/3.1)^4))$
 $= 642256.062 \text{ mm}^4$

Flange Inertia [Ip]:
 $= \max(Ip,term1, Ip,term2)$
 $= \max(689715, 642256)$
 $= 689714.6 \text{ mm}^4$

Flange Moment of Inertia for Integral Types [I]:
 $= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$
 $= 0.0874 \cdot 1.41 \cdot 4^2 \cdot 25.3 \cdot 160 / 0.014$
 $= 564323.500 \text{ mm}^4$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	46343.	43.2735	1.0000	2006248.
Face Pressure, Mt	23383.	45.7109	1.0000	1069269.
Gasket Load, Mg	55683.	36.6367	1.0000	2040880.
Gasket Seating, Matm	402105.	36.6367	1.0000	14737776.
Moment Load, Moe	...	...	...	301857.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 429150 [564324 / (0.384 \cdot 689715 + 564324)] \cdot (43.3 / (270 - 2 \cdot 43.3)) +$
 $598 \cdot 43.3$
 $= 301856.906 \text{ N-mm}$

Total Moment for Operation, Mo 5418254. N-mm
 Total Moment for Gasket seating, Mg 14737776. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 25.312 mm
 Hub Ratio, h/h0 = HL / H0 2.448
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 5.756

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.478
 Factor V per 4.16.5 0.014
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:
 $= A / B$
 $= 318/160$
 $= 1.982$

Stress factor from Table 4.16.4 [Y]:
 $= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.98^2 \cdot \log_{10}(1.98))) / (1.98^2 - 1)) / (1.98 - 1)$
 $= 3.001$

Stress factor from Table 4.16.4 [T]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / ((1.0472 + 1.9448 \cdot 1.98^2) \cdot (1.98 - 1))$
 $= 1.513$

Stress factor from Table 4.16.4 [U]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / (1.36136 (1.98^2 - 1) \cdot (1.98 - 1))$
 $= 3.297$

Stress factor from Table 4.16.4 [Z]:
 $= (K^2 + 1) / (K^2 - 1)$

$$= (1.98^2 + 1) / (1.98^2 - 1)$$

$$= 1.683$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.3 \cdot 4^2 \cdot 25.3 / 0.014$$

$$= 94269.492 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.48 / 25.3$$

$$= 0.019 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (32.1 \cdot 0.019 + 1) / 1.51 + 32.1^3 / 94269$$

$$= 1.410$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 5418254 / 160) / (1.41 \cdot 23^2)$$

$$= 45.25 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 14737776 / 160) / (1.41 \cdot 23^2)$$

$$= 123.07 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (1.81 \cdot 5418254 / 160) / (1.41 \cdot 32.1^2)$$

$$= 42.18 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (1.81 \cdot 14737776 / 160) / (1.41 \cdot 32.1^2)$$

$$= 114.73 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (3 \cdot 5418254 / (32.1^2 \cdot 160)) - 1.68 \cdot 42.2$$

$$= 27.79 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (Y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (3 \cdot 14737776 / (32.1^2 \cdot 160)) - 1.68 \cdot 115$$

$$= 75.58 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (45.2 + \max(42.2, 27.8)) / 2$$

$$= 43.71 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (123 + \max(115, 75.6)) / 2$$

$$= 118.90 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (W_{m1} + F_a + 4M_e/G) / A_b$$

$$= 134741 / 2338$$

$$= 57.63 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (278215 / 2338)$$

$$= 119.00 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	45.25	207.00	123.07	207.00
Radial Flange	42.18	138.00	114.73	138.00
Tangential Flange	27.79	138.00	75.58	138.00
Maximum Average	43.71	138.00	118.90	138.00
Bolting	57.63	172.00	119.00	172.00

Minimum Required Flange Thickness 32.969 mm

Estimated M.A.W.P. (Operating) 7.2 MPa
 Estimated Finished Weight of Flange at given Thk. 20.7 kgm
 Estimated Unfinished Weight of Forging at given Thk 45.5 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14925671 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.334$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5487333 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.125$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14737776 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.313$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5418254 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.117$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	39.05	207.00	103.57	207.00
Radial Flange	38.41	138.00	101.87	138.00
Tangential Flange	21.36	138.00	56.65	138.00
Maximum Average	38.73	138.00	102.72	138.00
Bolting	57.63	172.00	119.00	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 31.013 mm

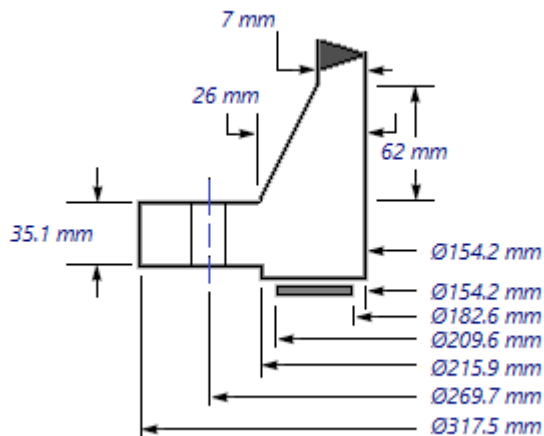
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Flange Input Data Values

Description: FLANGE-7 :

Flange-7

Flange Type	Integral Weld Neck		
Design Pressure	P	2.30	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	154.178	mm
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	35.0520	mm
Thickness of Hub at Small End	go	7.0000	mm
Thickness of Hub at Large End	gl	26.0230	mm
Length of Hub	h	61.9760	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	154.178	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		245.17	N
Bending Moment on Flange		12840.90	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.3 \cdot (154/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 2.3) + Ca$$

$$= 4.3483 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 4) / (80.1 + 0.6 \cdot 4)$$

$$= 6.692 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 7) / (77.1 + 0.6 \cdot 7)$$

$$= 11.883 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - ci$	32.052 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot ci$	160.178 mm
Corroded Large Hub, $g1_{cor} = g1 - ci$	23.023 mm
Corroded Small Hub, $g0_{cor} = go - ci$	4.000 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	31.762 mm
Gasket Contact Width, $N = (Go - Gi) / 2$	13.462 mm
Basic Gasket Width, $bo = N / 2$	6.731 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.538 mm
Gasket Reaction Diameter, $G = Go - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 196^2 \cdot 2.3$$

$$= 69725.781 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.3$$

$$= 55683.285 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 160^2 \cdot 2.3$$

$$= 46343.246 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 69726 - 46343$$

$$= 23382.537 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(69726 + 55683 + 0, 0)$$

$$= 125409.062 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((125409 + 245 + 4 \cdot 12841 / 196) / 172, 278215 / 172)$$

$$= 1617.694 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19 + 6 \cdot 32.1 / (3 + 0.5)$$

$$= 93.046 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 270 \cdot \sin(\pi/12)$$

$$= 69.816 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{69.8/(2 \cdot 19 + 32.1)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Hub and Bolts:	28.575	31.762	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	93.046

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi(210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 125409 - 69726$$

$$= 55683.28 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((270 - 160) / 2 + 36.6)$$

$$= 45.7109 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (270 - 160 - 23) / 2$$

$$= 43.2735 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 78.7, Bb = 32.1, Cc = 62 \text{ and } Ddg = 13.5 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 78.7 \cdot 32.1^3 \cdot (1/3 - 0.210 \cdot (32.1/78.7) \cdot (1 - 1/12 \cdot (32.1/78.7)^4))$$

$$= 642256.062 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 62 \cdot 13.5^3 \cdot (1/3 - 0.105 \cdot (13.5/62) \cdot (1 - 1/192 \cdot (13.5/62)^4))$$

$$= 47458.582 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 39.2 + 2.9$$

$$= 689714.625 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:



$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 78.7 \cdot 32.1^3 (1/3 - 0.21 (32.1/78.7) (1 - 1/12 (32.1/3.1)^4))$$

$$= 642256.062 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(I_p, \text{term1}, I_p, \text{term2})$$

$$= \max(689715, 642256)$$

$$= 689714.6 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot g_o^2 \cdot h_o \cdot B / V$$

$$= 0.0874 \cdot 1.41 \cdot 4^2 \cdot 25.3 \cdot 160 / 0.014$$

$$= 564323.500 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	46343.	43.2735	1.0000	2006248.
Face Pressure, Mt	23383.	45.7109	1.0000	1069269.
Gasket Load, Mg	55683.	36.6367	1.0000	2040880.
Gasket Seating, Matm	402105.	36.6367	1.0000	14737776.
Moment Load, Moe	...	...	...	18871.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 12841 [564324 / (0.384 \cdot 689715 + 564324)] \cdot (43.3 / (270 - 2 \cdot 43.3)) + 245 \cdot 43.3$$

$$= 18870.768 \text{ N-mm}$$

Total Moment for Operation, Mo 5135268. N-mm
 Total Moment for Gasket seating, Mg 14737776. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 25.312 mm
 Hub Ratio, h/h0 = HL / H0 2.448
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 5.756

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.478
 Factor V per 4.16.5 0.014
 Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 318 / 160$$

$$= 1.982$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.98^2 \cdot \log_{10}(1.98))) / (1.98^2 - 1)) / (1.98 - 1)$$

$$= 3.001$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / ((1.0472 + 1.9448 \cdot 1.98^2) \cdot (1.98 - 1))$$

$$= 1.513$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.98^2 (1 + 8.55246 \cdot \log_{10}(1.98)) - 1) / (1.36136 (1.98^2 - 1) \cdot (1.98 - 1))$$

$$= 3.297$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.98^2 + 1) / (1.98^2 - 1)$$

$$= 1.683$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 3.3 \cdot 4^2 \cdot 25.3 / 0.014$$

$$= 94269.492 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.48 / 25.3$$

$$= 0.019 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (32.1 \cdot 0.019 + 1) / 1.51 + 32.1^3 / 94269$$

$$= 1.410$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 5135268 / 160) / (1.41 \cdot 23^2)$$

$$= 42.88 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 14737776 / 160) / (1.41 \cdot 23^2)$$

$$= 123.07 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (1.81 \cdot 5135268 / 160) / (1.41 \cdot 32.1^2)$$

$$= 39.98 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (1.81 \cdot 14737776 / 160) / (1.41 \cdot 32.1^2)$$

$$= 114.73 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_o / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (3 \cdot 5135268 / (32.1^2 \cdot 160)) - 1.68 \cdot 40$$

$$= 26.33 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (3 \cdot 14737776 / (32.1^2 \cdot 160)) - 1.68 \cdot 115$$

$$= 75.58 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (42.9 + \max(40, 26.3)) / 2$$

$$= 41.43 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (123 + \max(115, 75.6)) / 2$$

$$= 118.90 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (W_{m1} + F_a + 4M_e/G) / A_b$$

$$= 125916 / 2338$$

$$= 53.86 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (278215 / 2338)$$

$$= 119.00 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	42.88	207.00	123.07	207.00
Radial Flange	39.98	138.00	114.73	138.00
Tangential Flange	26.33	138.00	75.58	138.00
Maximum Average	41.43	138.00	118.90	138.00
Bolting	53.86	172.00	119.00	172.00

Minimum Required Flange Thickness 32.969 mm

Estimated M.A.W.P. (Operating) 6.69 MPa

Estimated Finished Weight of Flange at given Thk. 20.7 kgm

Estimated Unfinished Weight of Forging at given Thk 45.5 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14925671 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.334$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5200738 / 1.01 \cdot 1 \cdot 0.014 / (1.32 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.118$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14737776 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 202713 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.313$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5135268 / 1 \cdot 1 \cdot 0.014 / (1.41 \cdot 199340 \cdot 4^2 \cdot 25.3 \cdot 0.3)$
 $= 0.111$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	36.89	207.00	103.57	207.00
Radial Flange	36.28	138.00	101.87	138.00
Tangential Flange	20.18	138.00	56.65	138.00
Maximum Average	36.59	138.00	102.72	138.00
Bolting	53.86	172.00	119.00	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 31.013 mm

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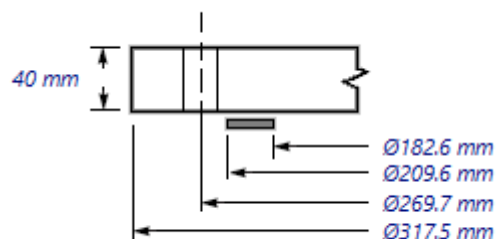
Flange Input Data Values

Description: FLANGE-8 :

Flange-8

Flange Type		Blind	
Design Pressure	P	2.30	MPa
Design Temperature		80	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	317.500	mm
Flange Thickness	t	40.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	847.438	mm
Diameter of Bolt Circle	C	269.748	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		12	
Flange Face Outside Diameter	Fod	215.900	mm
Flange Face Inside Diameter	Fid	133.350	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	209.550	mm
Gasket Inside Diameter	Gi	182.626	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		98.07	N
Bending Moment on Flange		1420.68	N-mm

External forces and moments are not considered in determining required bolt areas for bolted covers in Part 4.16 flange analysis, see Table 4.16.7.



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Gasket Contact Width,	$N = (Go - Gi)/2$	13.462 mm
Basic Gasket Width,	$bo = N/2$	6.731 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.538 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	196.475 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 196^2 \cdot 2.3$$

$$= 69725.781 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.54 \cdot \pi \cdot 196 \cdot 3 \cdot 2.3$$

$$= 55683.285 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(69726 + 55683 + 0, 0)$$

$$= 125409.062 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.54 \cdot \pi \cdot 196 + 0 \cdot 0$$

$$= 278214.656 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((125409 + 98.1 + 4 \cdot 1421/196) / 172, 278215/172)$$

$$= 1617.694 \text{ mm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1617.694	2338.060	
Radial Distance between Bolts and Edge:	20.637	23.876	
Circ. Spacing between the Bolts:	44.450	69.816	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 2338 \cdot 172 / (68.9 \cdot \pi (210 + 183))$$

$$= 4.734 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 2338$$

$$= 402104.81 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 125409.06 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (270 - 196) / 2$$

$$= 36.6367 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 125409 \cdot 36.6 \cdot 1 / (37^2 \cdot 196) + 0.3 \cdot 1 \cdot 2.3 \cdot 196^2 / 37^2$$

$$= 51.91 \text{ MPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 402105 \cdot 36.6 \cdot 1 / (37^2 \cdot 196)$$

$$= 104.07 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$



= 125536/2338
 = 53.70 MPa

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (278215/2338)
 = 119.00 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Tangential Flange	51.91	138.00	104.07	138.00
Bolting	53.70	172.00	119.00	172.00

Reqd. Blind Flange Thickness at Center 35.132 mm
 Reqd. Blind Flange Thickness at Gasket 35.132 mm

Estimated M.A.W.P. (Operating) 6.11 MPa
 Estimated Finished Weight of Flange at given Thk. 24.5 kgm
 Estimated Unfinished Weight of Forging at given Thk 24.5 kgm

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Tangential Flange	44.42	138.00	89.05	138.00
Bolting	53.70	172.00	119.00	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 32.156 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Bottom head		2.314	8.2353	3	168.3	118	1
Flange-1		2.3132	38	3.175	154.18	138	2
Flange-2		2.3122	38	3.175	154.18	138	3
Shell-1		2.3113	7	3	168.3	118	4
Flange -3		2.3	38	3	154.18	138	5
Flange-4		2.3	38	3	154.18	138	6
Shell-2		2.3	7	3	168.3	118	7
Flange-5		2.3	38	3	154.18	138	8
Flange-6		2.3	38	3	154.18	138	9
Shell-3		2.3	7	3	168.3	118	10
Flange-7		2.3	38	3	154.18	138	11
Flange-8		2.3	40	3	317.5	138	12

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Bottom head		2.3	6.13	10.6	7	4.5875	1
Flange-1		2.3	7.3	7.32	35.052	33.7312	2
Flange-2		2.3	6.38	7.31	35.052	33.0454	3
Shell-1		2.3	5.71	10.2	7	4.63547	4
Flange -3		2.3	6.64	6.64	35.052	32.9692	5
Flange-4		2.3	6.57	6.57	35.052	33.3248	6
Shell-2		2.3	5.72	10.2	7	4.62755	7
Flange-5		2.3	7.16	7.16	35.052	32.9692	8
Flange-6		2.3	7.2	7.2	35.052	32.9692	9
Shell-3		2.3	5.72	10.2	7	4.62755	10
Flange-7		2.3	6.69	7.37	35.052	32.9692	11
Flange-8		2.3	6.11	7.15	40	35.1318	12

Minimum 5.71 6.57

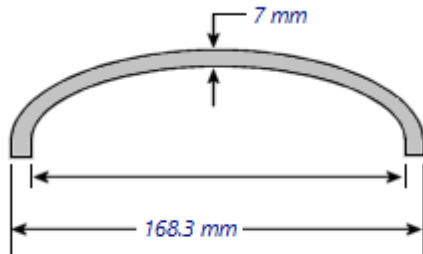
MAWP: 5.71 MPa, limited by: Shell-1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	2.314 MPa	Int. Design Temperature		80 °C
Material		SA-333 6	Ext. Design Temperature		80 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K03006
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	168.300 mm	Actual Stress		44.4 MPa
Spec. Min. Thk t _e , or sa		7.0000 mm	Spec. Nominal Thk.		8.235 mm
Surface Area		58436.4 mm ²	SG of Contents		1.000
Element Volume		1.4 ltr	Weight of Contents		13.9 N
Empty Weight		35.8 N	Operating Weight		49.6 N
			Comp. All. Ope. Stress		141.6 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 2.001 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.000 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

= $(P \cdot D_o \cdot K_{cor}) / (2 \cdot S \cdot E + 2 \cdot P \cdot (K_{cor} - 0.1))$ per Appendix 1-4 (c)

= $(2.31 \cdot 168 \cdot 0.95) / (2 \cdot 118 \cdot 1 + 2 \cdot 2.31 \cdot (0.95 - 0.1))$

= 1.5464 + 3.0000 = 4.5464 mm

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5875 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.014 MPa

= $(2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1))$ per Appendix 1-4 (c)

= $(2 \cdot 118 \cdot 1 \cdot 4) / (0.95 \cdot 168 - 2 \cdot 4 \cdot (0.95 - 0.1))$

= 6.15 - 0.014 = 6.13 MPa

Maximum Allowable Pressure, New and Cold [MAPNC]:

= $(2 \cdot S \cdot E \cdot t) / (K \cdot D_o - 2 \cdot t \cdot (K - 0.1))$ per Appendix 1-4 (c)

= $(2 \cdot 118 \cdot 1 \cdot 7) / (1 \cdot 168 - 2 \cdot 7 \cdot (1 - 0.1))$

= 10.6 MPa



Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1))) / (2 \cdot E \cdot t) \\ &= (2.31 \cdot (0.95 \cdot 168 - 2 \cdot 4 \cdot (0.95 - 0.1))) / (2 \cdot 1 \cdot 4) \\ &= 44.409 \text{ MPa} \end{aligned}$$

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) + c_a \text{ per Appendix 1-1 (a)(1)} \\ &= (2.31 \cdot 84.2) / (118 \cdot 1 + 0.4 \cdot 2.31) + 3 \\ &= 4.637 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.014 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)} \\ &= (118 \cdot 1 \cdot 5.24) / (84.2 - 0.4 \cdot 5.24) \\ &= 7.53 - 0.014 = 7.51 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (160 / (2 \cdot 41.6)))^2 / 6 \\ &= 0.952762 \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

MDMT Calculations in the Head Straight Flange:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 2.001 mm

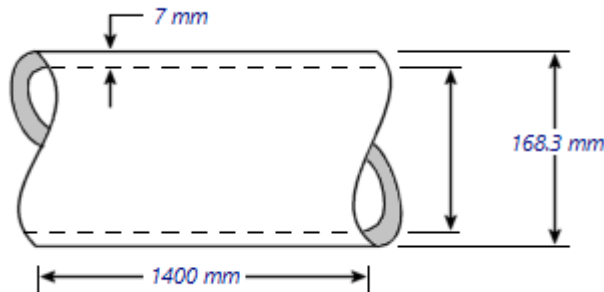
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.000 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	2.311 MPa	Int. Design Temperature		80 °C
Material		SA-333 6	Ext. Design Temperature		80 °C
Length		1400.000 mm	External Pressure Chart		CS-2
			UNS Number		K03006
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	168.300 mm	Actual Stress		47.7 MPa
Spec. Min. Thk t _e or s _a		7.0000 mm	Spec. Nominal Thk.		7.000 mm
Surface Area		740222.1 mm ²	SG of Contents		1.000
Element Volume		26.2 ltr	Weight of Contents		210.8 N
Empty Weight		377.4 N	Operating Weight		588.2 N
			Comp. All. Ope. Stress		141.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (2.31 \cdot 84.2) / (118 + 0.4 \cdot 2.31)$$

$$= 1.6355 + 3.0000 = 4.6355 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118 \cdot 1 \cdot 4) / (84.2 - 0.4 \cdot 4)$$

$$= 5.72 - 0.011 = 5.71 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118 \cdot 1 \cdot 7) / (84.2 - 0.4 \cdot 7)$$

$$= 10.2 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$$

$$= (2.31 \cdot (84.2 - 0.4 \cdot 4)) / (1 \cdot 4)$$

$$= 47.699 \text{ MPa}$$

Minimum Design Metal Temperature Results:

Note:



This Material was specified as being an Impact Tested (Low Temperature) Material.

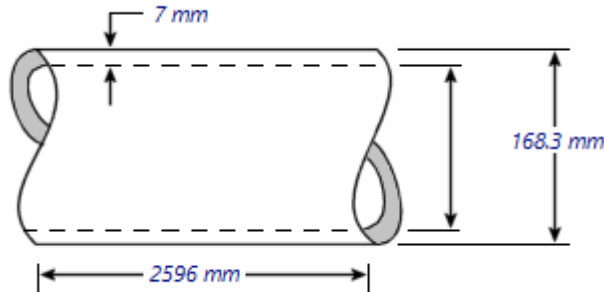
Impact Test Temperature provided per Specification -46 °C

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	2.300 MPa	Int. Design Temperature		80 °C
Material		SA-333 6	Ext. Design Temperature		80 °C
Length		2596.000 mm	External Pressure Chart		CS-2
			UNS Number		K03006
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	168.300 mm	Actual Stress		47.5 MPa
Spec. Min. Thk t _e or s _a		7.0000 mm	Spec. Nominal Thk.		7.000 mm
Surface Area		1372583.2 mm ²	SG of Contents		0.000
Element Volume		48.6 ltr	Weight of Contents		0.0 N
Empty Weight		699.9 N	Operating Weight		699.9 N
			Comp. All. Ope. Stress		141.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (2.3 \cdot 84.2) / (118 \cdot 1 + 0.4 \cdot 2.3)$$

$$= 1.6276 + 3.0000 = 4.6276 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118 \cdot 1 \cdot 4) / (84.2 - 0.4 \cdot 4)$$

$$= 5.72 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118 \cdot 1 \cdot 7) / (84.2 - 0.4 \cdot 7)$$

$$= 10.2 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (2.3 \cdot (84.2 - 0.4 \cdot 4)) / (1 \cdot 4)$$

$$= 47.466 \text{ MPa}$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



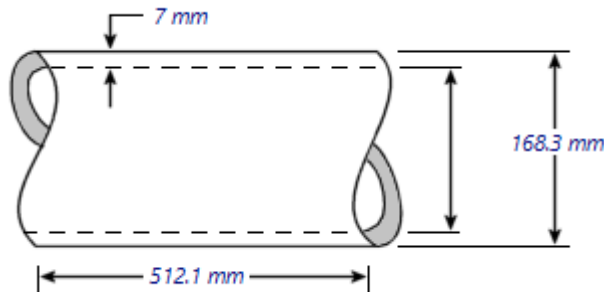
Impact Test Temperature provided per Specification -46 °C

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	2.300 MPa	Int. Design Temperature		80 °C
Material		SA-333 6	Ext. Design Temperature		80 °C
Length		512.100 mm	External Pressure Chart		CS-2
			UNS Number		K03006
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	168.300 mm	Actual Stress		47.5 MPa
Spec. Min. Thk t _e or s _a		7.0000 mm	Spec. Nominal Thk.		7.000 mm
Surface Area		270762.7 mm ²	SG of Contents		0.000
Element Volume		9.6 ltr	Weight of Contents		0.0 N
Empty Weight		138.1 N	Operating Weight		138.1 N
			Comp. All. Ope. Stress		141.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (2.3 \cdot 84.2) / (118 \cdot 1 + 0.4 \cdot 2.3)$$

$$= 1.6276 + 3.0000 = 4.6276 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118 \cdot 1 \cdot 4) / (84.2 - 0.4 \cdot 4)$$

$$= 5.72 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118 \cdot 1 \cdot 7) / (84.2 - 0.4 \cdot 7)$$

$$= 10.2 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (2.3 \cdot (84.2 - 0.4 \cdot 4)) / (1 \cdot 4)$$

$$= 47.466 \text{ MPa}$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -46 °C

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

UG99(b) = $1.30 * \text{M.A.W.P.} * \text{Sa/S}$ 7.418 MPa
 UG99b(b)(35) = $1.30 * \text{Design Pres} * \text{Sa/S}$ 2.990 MPa
 UG99(c) = $1.30 * \text{M.A.P.} - \text{Head(Hyd)}$ 8.535 MPa
 UG100 = $1.10 * \text{M.A.W.P.} * \text{Sa/S}$ 6.277 MPa
 PED = $\max(1.43*DP, 1.25*DP*ratio)$ 3.289 MPa
 PED&UG99(b)(35) = $\max(1.43*DP, 1.3*DP*ratio)$ 3.289 MPa
 App 27-4 = M.A.W.P. 5.706 MPa

User Defined Hydrostatic Test Pressure at High Point 3.050 MPa

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	33.9	216.0	0.157	3.05
Shell-1	35.5	216.0	0.164	3.05
Shell-2	35.5	216.0	0.164	3.05
Shell-3	35.5	216.0	0.164	3.05

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4 1"	Nozzle	138.00	138.00	1.000
N1 2"	Nozzle	138.00	138.00	1.000
K1A 1"	Nozzle	138.00	138.00	1.000
K1B 1"	Nozzle	138.00	138.00	1.000
K2A 1"	Nozzle	138.00	138.00	1.000
K2B 1"	Nozzle	138.00	138.00	1.000
K3A 1"	Nozzle	138.00	138.00	1.000
K3B 1"	Nozzle	138.00	138.00	1.000
K3C 1"	Nozzle	138.00	138.00	1.000
K3D 1"	Nozzle	138.00	138.00	1.000
K3E 1"	Nozzle	138.00	138.00	1.000
K3F 1"	Nozzle	138.00	138.00	1.000
K4B 1"	Nozzle	138.00	138.00	1.000
N2 1"	Nozzle	138.00	138.00	1.000
K4A 1"	Nozzle	138.00	138.00	1.000
N3 2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	118.00	118.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-1	118.00	118.00	1.000
Flange -3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-2	118.00	118.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-3	118.00	118.00	1.000
Flange-7	138.00	138.00	1.000
Flange-8	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4 1"	6.68	225.00	0.030
N1 2"	11.52	225.00	0.051
K1A 1"	4.56	225.00	0.020
K1B 1"	4.56	225.00	0.020
K2A 1"	4.56	225.00	0.020
K2B 1"	4.56	225.00	0.020
K3A 1"	4.56	225.00	0.020
K3B 1"	4.56	225.00	0.020
K3C 1"	4.56	225.00	0.020
K3D 1"	4.56	225.00	0.020
K3E 1"	4.56	225.00	0.020
K3F 1"	4.56	225.00	0.020
K4B 1"	4.56	225.00	0.020
N2 1"	6.68	225.00	0.030
K4A 1"	4.56	225.00	0.020
N3 2"	11.37	225.00	0.051

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.418 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.990 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	8.503 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	6.277 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.289 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.289 MPa
App 27-4	= M.A.W.P.	5.706 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 5.71 • 1
 = 7.418 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	143.4	216.0	0.664	7.47
Shell-1	154.1	216.0	0.713	7.47
Shell-2	153.8	216.0	0.712	7.45
Shell-3	153.2	216.0	0.709	7.42

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4 1"	Nozzle	138.00	138.00	1.000
N1 2"	Nozzle	138.00	138.00	1.000
K1A 1"	Nozzle	138.00	138.00	1.000
K1B 1"	Nozzle	138.00	138.00	1.000
K2A 1"	Nozzle	138.00	138.00	1.000
K2B 1"	Nozzle	138.00	138.00	1.000
K3A 1"	Nozzle	138.00	138.00	1.000
K3B 1"	Nozzle	138.00	138.00	1.000
K3C 1"	Nozzle	138.00	138.00	1.000
K3D 1"	Nozzle	138.00	138.00	1.000
K3E 1"	Nozzle	138.00	138.00	1.000
K3F 1"	Nozzle	138.00	138.00	1.000
K4B 1"	Nozzle	138.00	138.00	1.000
N2 1"	Nozzle	138.00	138.00	1.000
K4A 1"	Nozzle	138.00	138.00	1.000
N3 2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	118.00	118.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-1	118.00	118.00	1.000
Flange -3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-2	118.00	118.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-3	118.00	118.00	1.000
Flange-7	138.00	138.00	1.000



Flange-8	138.00	138.00	1.000
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Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4 1"	16.34	225.00	0.073
N1 2"	28.19	225.00	0.125
K1A 1"	11.15	225.00	0.050
K1B 1"	11.15	225.00	0.050
K2A 1"	11.15	225.00	0.050
K2B 1"	11.15	225.00	0.050
K3A 1"	11.15	225.00	0.050
K3B 1"	11.15	225.00	0.050
K3C 1"	11.15	225.00	0.050
K3D 1"	11.15	225.00	0.050
K3E 1"	11.15	225.00	0.050
K3F 1"	11.15	225.00	0.050
K4B 1"	11.15	225.00	0.050
N2 1"	16.24	225.00	0.072
K4A 1"	11.08	225.00	0.049
N3 2"	27.63	225.00	0.123

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	168.3	4	0.003301	113.543
20	30	No Calc	...	31.877	No Calc	No Calc
30	40	No Calc	...	31.877	No Calc	No Calc
40	50	1400	168.3	4	0.00066722	66.7075
50	60	No Calc	...	32.052	No Calc	No Calc
60	70	No Calc	...	32.052	No Calc	No Calc
70	80	2596	168.3	4	0.00063425	63.4107
80	90	No Calc	...	32.052	No Calc	No Calc
90	100	No Calc	...	32.052	No Calc	No Calc
100	110	512.1	168.3	4	0.0014805	96.4495
110	120	No Calc	...	32.052	No Calc	No Calc
120	130	No Calc	...	37	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	7	No Calc	...	3
20	30	35.052	33.7312	...	No Calc
30	40	35.052	33.0454	...	No Calc
40	50	7	No Calc	...	2.11
50	60	35.052	32.9692	...	No Calc
60	70	35.052	33.3248	...	No Calc
70	80	7	No Calc	...	2.01
80	90	35.052	32.9692	...	No Calc
90	100	35.052	32.9692	...	No Calc
100	110	7	No Calc	...	3.06
110	120	35.052	32.9692	...	No Calc
120	130	40	35.1318	...	No Calc

Minimum

2.01

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	1400	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	2596	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	512.1	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	3.64769	1.41609	2.31889	1.54855	0.25534
20	30	22.5945	1.9265	21.3783	2.02379	1.58162
30	40	22.0724	1.9265	20.8563	2.02379	1.54507
40	50	38.4893	26.1835	22.403	28.2594	2.69425
50	60	22.0724	1.92315	20.9246	2.01497	1.54507
60	70	22.4172	1.92315	21.2693	2.01497	1.5692
70	80	71.3702	48.5517	41.5415	52.401	4.99592
80	90	22.6435	1.92315	21.4956	2.01497	1.58504
90	100	22.6435	1.92315	21.4956	2.01497	1.58504
100	110	14.0788	9.57755	8.19469	10.3369	0.98552
110	120	22.0724	1.92315	20.9246	2.01497	1.54507
120	130	24.5452	...	24.5452	...	1.71817
Total		308	99.20	247	106.67	21

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	1.41523	...	-19.2875	...	Liquid 10
10	Insl	3.17624	...	5.95	...	Ins: 10
10	Nozl	8.52683	...	-187.462	...	N4 1"
20	Liqd	1.81069	...	48.514	...	Liquid 20
20	Insl	4.27937	...	48.514	...	Ins: 20
30	Liqd	1.81069	...	48.514	...	Liquid 30
30	Insl	4.1805	...	48.514	...	Ins: 30
40	Liqd	21.4948	...	575	...	Liquid 40
40	Insl	33.5146	...	700	...	Ins: 40
40	Nozl	11.2272	-175.575	799.999	...	N1 2"
40	Nozl	5.99456	203.147	1102.77	...	K1A 1"
40	Nozl	5.99456	203.147	152.769	...	K1B 1"
40	Nozl	5.99456	...	1102.77	-203.147	K2A 1"
40	Nozl	5.99456	...	152.769	-203.147	K2B 1"
40	Nozl	5.99456	-203.147	1102.77	...	K3A 1"
40	Nozl	5.99456	-203.147	152.769	...	K3B 1"
40	Nozl	5.99456	...	1102.77	203.147	K3C 1"
40	Nozl	5.99456	...	152.769	203.147	K3D 1"
40	Nozl	5.99456	143.646	1102.77	-143.646	K3E 1"
40	Nozl	6.15957	143.646	152.769	-143.646	K3F 1"
40	Nozl	5.99456	-203.147	1350	...	K4B 1"
50	Insl	4.1805	...	48.514	...	Ins: 50
60	Insl	4.24579	...	48.514	...	Ins: 60
60	Wght	2	...	77.026	...	Bed sup ring
70	Pack	11.7007	...	1250	...	Bed
70	Insl	62.1457	...	1298	...	Ins: 70
70	Lugs	15.2559	...	1255	...	Support Lugs
70	Wght	100	...	1500	...	Cleat weight
70	Wght	1	...	705.538	...	Bed limiter
70	Wght	1.5	...	-0.0018898	...	Bed sup grid
80	Insl	4.28865	...	48.514	...	Ins: 80
90	Insl	4.28865	...	48.514	...	Ins: 90
100	Pack	0.16849	...	388	...	Demister
100	Insl	12.2592	...	256.05	...	Ins: 100
100	Nozl	10.648	202.156	162.307	...	N2 1"
100	Nozl	5.99456	-203.147	162.307	...	K4A 1"
100	Wght	0.45	...	162.307	...	N2 Feed pipe
100	Wght	1	...	159.996	...	Distributor
100	Wght	1	...	149.996	...	Dist sup
100	Wght	10	80	312.307	...	Lifting lug 1
100	Wght	10	-80	312.307	...	Lifting lug 2
100	Wght	500	100	312.307	...	Pipe davit



110	Ins1	4.1805	...	48.514	...
120	Ins1	3.24121	...	19	...
120	Noz1	11.8725	...	115.006	...
70	Pliq	4.67	...	1250.000	...

Ins: 110
 Ins: 120
 N3 2"

Total Weight of Each Detail Type:

Packing	11.9
Packing Liquid	4.7
Liquid	26.5
Insulation	144.0
Nozzles	114.4
Lugs	15.3
Weights	627.0
Sum of the Detail Weights	943.6 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	330.3	330.3	330.3	330.3	330.3	330.3
Nozzles	114.4	114.4	114.4	114.4	114.4	114.4
Lugs	15.3	15.3	15.3	15.3	15.3	15.3
Wld Weights	623.0	623.0	623.0	623.0	623.0	623.0
Packing	...	...	...	...	11.9	11.9
Insulation	...	...	...	144.0	144.0	144.0
Empty Weights	...	...	...	3.9	3.9	...
Test Weights	...	3.9	...	...	...	...
Ope Weights	...	...	...	...	...	3.9
Ope. Liquid	...	...	...	...	...	26.5
Packing Liq	...	...	...	...	...	4.7
Test Liquid	...	99.1	...	...	...	...
Totals	1082.9	1186.0	1082.9	1230.8	1242.7	1273.9

Field Installation Options:

* Packing installed after lifting.
 Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	1082.9 kgm
Shop Test Wt.	- Fabricated weight + water (full)	1186.0 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	1082.9 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1230.8 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1242.7 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	1273.9 kgm
Field Test Wt.	- Empty weight + test medium (full)	1330.0 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	725.7 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	752.3 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	745.4 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	58436.4	50045.4



20	30	139504	48424.8
30	40	138559	48424.8
40	50	740222	678647
50	60	138559	48424.8
60	70	139504	48424.8
70	80	1372583	1258405
80	90	138559	48424.8
90	100	138559	48424.8
100	110	270763	248239
110	120	138559	48424.8
120	130	119071	...

Total 3532879.2 2574310.5 mm²
 Total 3.5 2.6 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	12.4299	17.0213	17.0213	...	15.6061
20	30	24.1761	30.2662	30.3808	...	28.4555
30	40	23.6175	29.6087	29.7233	...	27.798
40	50	118.516	173.525	178.198	14156.7	152.031
50	60	23.6175	27.798	29.72	...	27.798
60	70	25.9864	30.2322	32.1542	...	30.2322
70	Lugs	81.3028	116.993	131.606	...	114.869
Lugs	80	97.5634	140.392	157.928	...	137.843
80	90	24.2285	28.5172	30.4391	...	28.5172
90	100	24.2285	28.5172	30.4391	...	28.5172
100	110	554.157	566.585	575.988	499669	566.585
110	120	23.6175	27.798	29.72	...	27.798
120	130	38.1359	41.3771	41.3771	...	41.3771

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	...	...	...
20	30	-15.6061	-17.0213	-17.0213
30	40	-44.0616	-47.2875	-47.4021
40	50	-71.8596	-76.8962	-77.1255
50	60	-223.89	-250.421	-255.324
60	70	-251.688	-278.22	-285.044
70	Lugs	-281.92	-308.452	-317.198
Lugs	80	830.637	833.186	865.891
80	90	692.794	692.794	707.963
90	100	664.277	664.277	677.524
100	110	635.76	635.76	647.085
110	120	69.1751	69.1751	71.0971
120	130	41.3771	41.3771	41.3771

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	14156.7	14156.7	14156.7



50	60	14156.7	14156.7	14156.7
60	70	14156.7	14156.7	14156.7
70	Lugs	14156.7	14156.7	14156.7
Lugs	80	499669	499669	499669
80	90	499669	499669	499669
90	100	499669	499669	499669
100	110	499669	499669	499669
110	120	...	...	...
120	130	...	...	...

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N4 1"	14.34	15.31	80	900	GR 1.1	30.434	14.338	...	-8.928
N1 2"	14.34	15.31	80	900	GR 1.1	5.230	14.338	14.338	14.338
K1A 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K1B 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K2A 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K2B 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K3A 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K3B 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K3C 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K3D 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K3E 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K3F 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
K4B 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
N2 1"	14.34	15.31	80	900	GR 1.1	30.434	14.338	...	-8.928
K4A 1"	9.58	10.20	80	600	GR 1.1	...	...	...	...
N3 2"	14.34	15.31	80	900	GR 1.1	...	...	...	...
Min Rating	9.576	10.205, [for Core Elements]					...	...	-8.928

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N4 1"	39.751	3.90	360555	1500	5.706	14.338	Pass
N1 2"	72.263	2.70	360555	1500	5.706	14.338	Pass
N2 1"	39.751	2.70	360555	1500	5.706	14.338	Pass

Evaluating Flange on Nozzle [N4 1"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(360555) + 4 \cdot \text{abs}(1500) \cdot 39.751 \leq \pi \cdot 39.751^3 \cdot ((14.338 - 5.706) + 3.900 \cdot 14.338)$$

$$= 0.60075E+07 \leq 0.12741E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4 1" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(14.338, 3.900 \cdot 14.338 + 14.338 - 6007482.5 / (\pi \cdot 39.751^3))$$

$$= \min(14.338, 39.820)$$

$$= 14.338 \text{ MPa}$$

Evaluating Flange on Nozzle [N1 2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(360555) + 4 \cdot \text{abs}(1500) \cdot 72.263 \leq \pi \cdot 72.263^3 \cdot ((14.338 - 5.706) + 2.700 \cdot 14.338)$$

$$= 0.62026E+07 \leq 0.56143E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(14.338, 2.700 \cdot 14.338 + 14.338 - 6202631.0 / (\pi \cdot 72.263^3))$$

$$= \min(14.338, 47.819)$$

$$= 14.338 \text{ MPa}$$

Evaluating Flange on Nozzle [N2 1"]:



$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(360555) + 4 \cdot \text{abs}(1500) 39.751 \leq \pi \cdot 39.751^3 [(14.338 - 5.706) + 2.700 \cdot 14.338] \\ &= 0.60075E+07 \leq 0.93452E+07 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2 1" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(14.338, 2.700 \cdot 14.338 + 14.338 - 6007482.5 / (\pi \cdot 39.751^3)) \\ &= \min(14.338, 22.615) \\ &= 14.338 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 4.61083 Hz.

The Natural Frequency for the Vessel (Ope...) is 4.60919 Hz.

The Natural Frequency for the Vessel (Filled) is 4.58175 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	4000	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 13.307

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 4.609 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (13.307) is > 5 .

Note:

As per Section 9.1, Dynamic effects of wind should be considered. The Cross-Wind dynamic effects can be considered using the Vortex Shedding which can be invoked from Design Constraints tab. While the Along-Wind effects can be considered using Gust Response Factor method, which can be invoked from the Wind Load tab.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_{2,1} \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50 \cdot 1 \cdot 1 \cdot 1 \cdot 1.15$$

$$= 57.496 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1984, 1 \cdot 1 \cdot 1 \cdot 0.6 \cdot 57.5^2)$$

$$= 1983.5 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 0.11 \cdot 0.93 \cdot 1984$$

$$= 194.6 \text{ N}$$

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	4121.76	1.0000	1.0000	1.0000	57.50	0.9331
20	4140.26	1.0000	1.0000	1.0000	57.50	0.9331
30	4237.29	1.0000	1.0000	1.0000	57.50	0.9331
40	4985.81	1.0000	1.0000	1.0000	57.50	0.9331
50	5737.50	1.0000	1.0000	1.0000	57.50	0.9331
60	5834.52	1.0000	1.0000	1.0000	57.50	0.9331
70	7181.04	1.0000	1.0000	1.0000	57.50	0.9331
80	8530.73	1.0000	1.0000	1.0000	57.50	0.9331
90	8627.75	1.0000	1.0000	1.0000	57.50	0.9331
100	8932.32	1.0000	1.0000	1.0000	57.50	0.9331
110	9240.06	1.0000	1.0000	1.0000	57.50	0.9331

120 9308.57 1.0000 1.0000 1.0000 57.50 0.9331

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Un corroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 DL - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.31416E-05$ (High); $0.31416E-05 < 0.39270E-05$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0/5290 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[1.33/17.4^2] \cdot 10^4 = 44.3$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.39270E-05$ no chance. [Vp]:

= $W / (L \cdot Dr^2)$

= $11849 / (5290 \cdot 338^2)$

= $0.19611E-04$

Since V_p is $> 0.39270E-05$ no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Bed sup ring	0.00	5824.46	1.98	0.00
Cleat weight	0.00	7344.46	1.98	0.00
Bed limiter	0.00	6550.00	1.98	0.00
Bed sup grid	0.00	5844.46	1.98	0.00
N2 Feed pipe	0.00	8800.00	1.98	0.00
Distributor	0.00	8797.69	1.98	0.00
Dist sup	0.00	8787.69	1.98	0.00
Lifting lug 1	0.00	8950.00	1.98	0.00
Lifting lug 2	0.00	8950.00	1.98	0.00
Pipe davit	0.00	8950.00	1.98	0.00

The Natural Frequency for the Vessel (Ope...) is 4.60919 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	4121.76	583.869	105121	1.98426	214.008
20	30	4140.26	563.674	54692.2	1.98426	111.344
30	40	4237.29	563.674	54692.2	1.98426	111.344



40	50	4985.81	583.869	817416	1.98426	1664.12
50	60	5737.5	563.674	54692.2	1.98426	111.344
60	70	5834.52	563.674	54692.2	1.98426	111.344
70	80	7181.04	583.869	1515723	1.98426	3085.75
80	90	8530.73	563.674	54692.2	1.98426	111.344
90	100	8627.75	563.674	54692.2	1.98426	111.344
100	110	8932.32	583.869	298999	1.98426	608.711
110	120	9240.06	563.674	54692.2	1.98426	111.344
120	130	9308.57	780.065	31202.6	1.98426	63.5231

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.2170 Seconds
Value of Sa/g from user Spectrum Data 0.1750
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.17 \cdot 1) \cdot 1$
 $= 0.175$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.22 secs
Computed Sa/g = 0.17

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 167 \cdot 0.17 \cdot 1$
 $= 29.2 \text{ N}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.17 \cdot 12342 \cdot 1$
 $= 1439.9 \text{ N}$

The Natural Frequency for the Vessel (Ope...) is 4.60919 Hz.

Earthquake Load Calculation:



From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	25	166.911	29.2094	21.3303	-19.473
20	30	101.689	296.79	51.9383	41.4874	-54.0985
30	40	198.717	290.343	50.81	40.5288	-87.9718
40	50	947.231	1701.59	297.778	203.379	-286.491
50	60	1698.92	272.587	47.7028	40.5288	-318.292
60	70	1795.95	296.457	51.8799	44.5939	-352.879
70	Lugs	3024.46	1147.24	200.766	139.52	-486.723
Lugs	80	3732.46	1376.68	240.92	167.424	953.192
80	90	4492.15	279.639	48.9369	41.5773	792.579
90	100	4589.18	279.639	48.9369	41.5773	759.955
100	110	4893.74	5555.93	972.287	950.961	727.33
110	120	5201.48	272.587	47.7028	40.5288	79.1386
120	130	5270	405.744	71.0052	65.4431	47.3368



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	3068.28	...	...	...	...
20	30	2922.77	214.008	29.2094	20085.1	2741.36
30	40	2825.75	325.352	81.1477	46942	8190.18
40	50	2077.23	436.696	131.958	83927	18533
50	60	1325.54	2100.81	429.736	1860904	411878
60	70	1228.51	2212.16	477.438	2076901	457271
70	Lugs	590	2323.5	529.318	2297033	506133
Lugs	80	708	6415.52	2159.87	2814327	1151230
80	90	1467.69	1006.27	1188.87	432640	542175
90	100	1564.72	894.922	1139.93	340369	429150
100	110	1869.28	783.578	1090.99	258906	320875
110	120	2177.02	174.867	118.708	12840.9	10628.1
120	130	2245.53	63.5231	71.0052	1270.97	1420.68

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	11.8961	12.0939	0.0029908
20	30	214.008	11.3146	11.6983	0.0029899
30	40	325.352	10.5471	10.9309	0.0029896
40	50	436.696	5.03304	10.1634	0.002989
50	60	2100.81	0.71748	0.95722	0.24307E-04
60	70	2212.16	0.23886	0.47802	0.12751E-04
70	Lugs	2323.5	-2.40475	...	...
Lugs	80	6415.52	0.93183	6.01356	-0.0024525
80	90	1006.27	6.37201	6.7305	-0.0024504
90	100	894.922	7.08904	7.44761	-0.0024488
100	110	783.578	9.34488	11.247	-0.0024084
110	120	174.867	11.6076	11.9681	-0.0024084
120	130	63.5231	12.1168	12.2654	-0.0024084

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	48.3104	301.94
20	30	46.6395	291.497
30	40	46.6395	291.497
40	50	48.3104	301.94
50	60	46.6395	291.497
60	70	46.6395	291.497
70	80	48.3104	301.94
80	90	46.6395	291.497
90	100	46.6395	291.497
100	110	48.3104	301.94
110	120	46.6395	291.497
120	130	64.544	403.4

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 26.450 Actual deflection : 12.265 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	3547.17	2064.66	137349	82839.2
20	30	3547.17	1976.43	137349	79464.3
30	40	3547.17	1976.43	137349	79464.3
40	50	3547.17	2064.66	137349	82839.2
50	60	3547.17	2064.66	137349	82839.2
60	70	3547.17	2064.66	137349	82839.2
70	80	3547.17	2064.66	137349	82839.2
80	90	3547.17	2064.66	137349	82839.2
90	100	3547.17	2064.66	137349	82839.2
100	110	3547.17	2064.66	137349	82839.2
110	120	3547.17	2064.66	137349	82839.2
120	130	3547.17	2064.66	137349	82839.2



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	141.597	259.195	-141.597	-145.465
20	30	165.597	269.995	-144.937	-144.937
30	40	165.597	269.995	-144.937	-144.937
40	50	141.597	259.195	-141.597	-145.465
50	60	165.597	269.995	-145.465	-145.465
60	70	165.597	269.995	-145.465	-145.465
70	Lugs	141.597	259.195	-141.597	-145.465
Lugs	80	141.597	259.195	-141.597	-145.465
80	90	165.597	269.995	-145.465	-145.465
90	100	165.597	269.995	-145.465	-145.465
100	110	141.597	259.195	-141.597	-145.465
110	120	165.597	269.995	-145.465	-145.465
120	130	165.597	269.995	-145.465	-145.465

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	22.5831	...	72.8382
20	30	...	...	...
30	40	...	...	...
40	50	22.5831	...	72.8382
50	60	...	...	...
60	70	...	...	...
70	80	22.5831	...	72.8382
80	90	...	...	...
90	100	...	...	...
100	110	22.5831	...	72.8382
110	120	...	...	...
120	130	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	...	...	...	...	...
20	30	0.077436	0.084458	0.084458	...	...
30	40	0.21863	0.23464	0.2352	...	...
40	50	0.34132	0.36525	0.36634	0.17084	0.17084
50	60	1.06345	1.18947	1.21275	0.17084	0.17084
60	70	1.19548	1.3215	1.35392	0.17084	0.17084
70	Lugs	1.33908	1.4651	1.50665	0.17084	0.17084
Lugs	80	-3.94541	-3.95752	-3.94541	6.02986	6.02986
80	90	-3.29068	-3.29068	-3.29068	6.02986	6.02986
90	100	-3.15522	-3.15522	-3.15522	6.02986	6.02986
100	110	-3.01977	-3.01977	-3.01977	6.02986	6.02986
110	120	-0.32857	-0.32857	-0.32857	...	...
120	130	-0.19654	-0.19654	-0.19654	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	...	...	...	...	...
20	30	...	0.25268	0.034487	0.083383	...
30	40	...	0.59054	0.10303	0.19488	...
40	50	0.17084	1.01281	0.22365	0.33423	...
50	60	0.17084	22.4569	4.97042	7.41076	...
60	70	0.17084	25.0635	5.51822	8.27094	...
70	Lugs	0.17084	27.7199	6.10786	9.14758	...
Lugs	80	6.02986	33.9625	13.8927	11.2076	...
80	90	6.02986	5.22098	6.54281	1.72292	...
90	100	6.02986	4.10747	5.17885	1.35547	...
100	110	6.02986	3.1244	3.87223	1.03105	...
110	120	...	0.15496	0.12826	0.051137	...
120	130	...	0.015338	0.017144	0.0050615	...



Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	...
20	30	...	...	...	0.025184
30	40	...	...	...	0.077414
40	50	...	...	...	0.17158
50	60	...	...	...	3.63703
60	70	...	...	...	4.03181
70	Lugs	...	...	...	4.4647
Lugs	80	...	...	...	16.2122
80	90	...	...	...	6.28856
90	100	...	...	...	4.97743
100	110	...	...	...	3.71502
110	120	...	...	...	0.1162
120	130	...	...	...	0.015801

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	-0.0094324
20	30	...	-0.027374
30	40	...	-0.044514
40	50	...	-0.13877
50	60	...	-0.15418
60	70	...	-0.17093
70	Lugs	...	-0.23576
Lugs	80	...	0.46171
80	90	...	0.38391
90	100	...	0.36811
100	110	...	0.35231
110	120	...	0.038333
120	130	...	0.022929

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	Lugs	...	...	...	...
Lugs	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	120	...	...	...	...
120	130	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
------	----	--------------------------



10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	Lugs	...
Lugs	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	141.60	0.00	141.60	0.0000	-0.0000
40	1.52	141.60	-0.84	141.60	0.0108	0.0059
70	29.23	141.60	-26.55	141.60	0.2064	0.1875
70	36.05	141.60	-43.94	141.60	0.2546	0.3103
100	6.13	141.60	-12.17	141.60	0.0433	0.0860

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		141.60	-0.01	141.60		0.0001
40	0.54	141.60	-0.14	141.60	0.0038	0.0010
70	5.74	141.60	-3.53	141.60	0.0405	0.0249
70	18.76	141.60	-25.73	141.60	0.1325	0.1817
100	7.08	141.60	-12.41	141.60	0.0500	0.0877

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	141.60	0.00	141.60	0.0000	-0.0000
40	1.55	141.60	-0.82	141.60	0.0109	0.0058
70	29.36	141.60	-26.43	141.60	0.2073	0.1866
70	36.03	141.60	-43.95	141.60	0.2545	0.3104



100 6.13 141.60 -12.17 141.60 0.0433 0.0860

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		141.60	-0.01	141.60		0.0001
40	0.62	141.60	-0.17	141.60	0.0044	0.0012
70	7.51	141.60	-5.05	141.60	0.0530	0.0357
70	16.43	141.60	-23.42	141.60	0.1160	0.1654
100	7.23	141.60	-12.57	141.60	0.0511	0.0888

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	259.20	0.00	145.46	0.0000	-0.0000
40	0.87	259.20	-0.14	145.46	0.0034	0.0010
70	10.83	259.20	-7.81	145.46	0.0418	0.0537
70	13.29	259.20	-21.18	145.46	0.0513	0.1456
100	4.04	259.20	-10.08	145.46	0.0156	0.0693

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	259.20	0.00	145.46	0.0000	-0.0000
40	0.54	259.20		145.46	0.0021	
70	1.68	259.20		145.46	0.0065	
70	2.08	259.20	-9.98	145.46	0.0080	0.0686
100	3.01	259.20	-9.05	145.46	0.0116	0.0622

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.58	141.60		141.60	0.1595	
40	24.13	141.60		141.60	0.1704	
70	51.94	141.60	-3.84	141.60	0.3668	0.0271
70	58.62	141.60	-21.37	141.60	0.4140	0.1509
100	28.72	141.60		141.60	0.2028	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.57	141.60		141.60	0.1594	
40	23.20	141.60		141.60	0.1639	
70	30.09	141.60		141.60	0.2125	
70	39.01	141.60	-0.84	141.60	0.2755	0.0059
100	29.82	141.60		141.60	0.2106	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	141.60	0.00	141.60	0.0000	-0.0000
40	1.55	141.60	-0.82	141.60	0.0109	0.0058
70	29.36	141.60	-26.43	141.60	0.2073	0.1866
70	36.03	141.60	-43.95	141.60	0.2545	0.3104
100	6.13	141.60	-12.17	141.60	0.0433	0.0860

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		141.60	-0.01	141.60		0.0001
40	0.62	141.60	-0.17	141.60	0.0044	0.0012
70	7.51	141.60	-5.05	141.60	0.0530	0.0357
70	16.43	141.60	-23.42	141.60	0.1160	0.1654



100 7.23 141.60 -12.57 141.60 0.0511 0.0888

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	72.84	259.20		145.46	0.2810	
40	73.71	259.20		145.46	0.2844	
70	83.66	259.20		145.46	0.3228	
70	86.13	259.20		145.46	0.3323	
100	76.88	259.20		145.46	0.2966	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	72.84	259.20		145.46	0.2810	
40	73.38	259.20		145.46	0.2831	
70	74.52	259.20		145.46	0.2875	
70	74.92	259.20		145.46	0.2891	
100	75.85	259.20		145.46	0.2926	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.58	141.60		141.60	0.1595	
40	23.10	141.60		141.60	0.1631	
70	24.09	141.60		141.60	0.1702	
70	24.67	141.60		141.60	0.1742	
100	25.59	141.60		141.60	0.1807	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.58	141.60		141.60	0.1595	
40	22.80	141.60		141.60	0.1610	
70	23.44	141.60		141.60	0.1655	
70	20.28	141.60		141.60	0.1432	
100	20.83	141.60		141.60	0.1471	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.58	141.60		141.60	0.1595	
40	23.12	141.60		141.60	0.1633	
70	24.22	141.60		141.60	0.1710	
70	24.66	141.60		141.60	0.1741	
100	25.59	141.60		141.60	0.1807	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.58	141.60		141.60	0.1595	
40	23.10	141.60		141.60	0.1631	
70	24.09	141.60		141.60	0.1702	
70	24.67	141.60		141.60	0.1742	
100	25.59	141.60		141.60	0.1807	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	141.60	0.00	141.60	0.0000	-0.0000
40	0.54	141.60		141.60	0.0038	
70	1.64	141.60		141.60	0.0116	
70	2.07	141.60	-9.99	141.60	0.0146	0.0705



100 3.01 141.60 -9.05 141.60 0.0213 0.0639

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.57	118.00		118.00	0.1913	
40	22.98	118.00		118.00	0.1948	
70	23.98	118.00		118.00	0.2033	
70	25.12	118.00		118.00	0.2129	
100	25.95	118.00		118.00	0.2199	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		118.00	-0.01	118.00		0.0001
40	0.40	118.00		118.00	0.0034	
70	1.40	118.00		118.00	0.0119	
70	2.53	118.00	-9.53	118.00	0.0215	0.0807
100	3.36	118.00	-8.70	118.00	0.0285	0.0737

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	22.58	141.60		141.60	0.1595	
40	23.12	141.60		141.60	0.1633	
70	24.22	141.60		141.60	0.1710	
70	24.66	141.60		141.60	0.1741	
100	25.59	141.60		141.60	0.1807	

Absolute Maximum of the all of the Stress Ratio's 0.4140

Governing Element: Shell-2

Governing Load Case 7 : IP+OW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom head	22.6	...	...	...	22.6	+ Tens
Bottom head	22.6	...	...	...	...	- Comp
Shell-1	22.6	0.4	0.2	1.0	24.1	+ Tens
Shell-1	22.6	0.4	-0.2	-1.0	...	- Comp
Shell-2	22.6	1.5	0.2	27.7	51.9	+ Tens
Shell-2	22.6	1.5	-0.2	-27.7	-3.8	- Comp
Shell-2	22.6	-4.0	6.0	34.0	58.6	+ Tens
Shell-2	22.6	-4.0	-6.0	-34.0	-21.4	- Comp
Shell-3	22.6	-3.0	6.0	3.1	28.7	+ Tens
Shell-3	22.6	-3.0	-6.0	-3.1	...	- Comp

Shop/Field Installation Options :

Packing is installed in the Field after being lifted.
 Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	3121.877 mm
Center of Gravity of the Liquid	685.615 mm
Center of Gravity of the Insulation	2644.876 mm
Center of Gravity of the Nozzles	1887.070 mm
Center of Gravity of the Lugs	3099.462 mm
Center of Gravity of the Added Weights (Operating)	4672.073 mm
Center of Gravity of the Added Weights (Empty)	4672.073 mm
Center of Gravity of Bare Shell New and Cold	2799.000 mm
Center of Gravity of Bare Shell Corroded	2851.665 mm
Vessel CG in the Operating Condition	3641.552 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3787.751 mm
Vessel CG in the Test Condition	3689.400 mm

Warning: Lifting Lug is Higher than top of vessel!

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	5277.30 mm
Total vessel weight (No Liquid) Twt	12069.35 N
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	18104.03 N
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	5468.00 mm
Design Reaction force at the tailing lug	5563.15 N
Design Reaction force at the lifting lug	12540.88 N
CG distance during field lifting operation	3787.75 mm
CG distance from tailing lug	3787.75 mm
CG distance from the nearer lifting lug/trunnion	1680.25 mm
Compressive Allowable Increase Factor	1.00

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	61.40	3395.71	118.00 (UG-23)
Shear	3.29	5217.89	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	0.0	0.0	5547.5	1.6
166.43	882357.9	6.4	5038.5	1.4
1682.87	6598657.5	48.1	2592.3	0.7
3395.71	8435869.0	61.4	16.8	0.0
4608.76	7454145.5	54.3	2036.7	0.6
5237.30	2843832.8	5.0	11741.8	1.6

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.52 at elevation 3396 mm - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

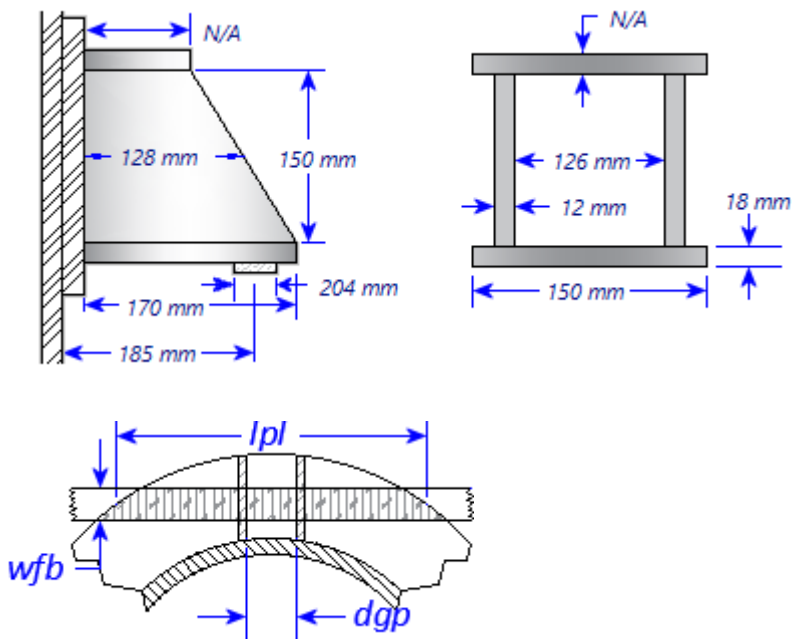
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Support Lug Calculations: Operating Case

Input Data: Support Lugs

Type of Geometry : Gussets with No Top Plate
 Number of Support Lugs Nlug 2
 Distance from Vessel OD to Lug Contact Point Dlug 185.0000 mm
 Lug Support Force Bearing Width Wfb 204.0000 mm
 Lug Material SA-516 70 +IT
 Lug Yield Stress 243.00 MPa
 Radial Width of bottom Support Lug Plate Wpl 170.0000 mm
 Effective Force Bearing Length Lpl 150.0000 mm
 Thickness of bottom Support Lug Plate Tpl 18.0000 mm
 Distance between Gussets Dgp 126.0000 mm
 Mean Width of Gusset Plate Wgp 128.0000 mm
 Height of Gusset Plate Hgp 150.0000 mm
 Thickness of Gusset Plate Tgp 12.0000 mm

Bolt Material SA-193 B7
 Bolt Allowable Stress at Design Temperature 172.00 MPa
 Thread Series TEMA Metric
 Bolt Diameter 24.00 mm



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug 3304825.5 N-mm
 Weight Load at the top of one Lug 6171.1 N

Force on one Lug, Operating Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$

$$= (12342/2 + 3304825/(269 \cdot 2/2))$$

$$= 18444.84 \text{ N}$$

Force on one lug in the Empty Condition [Flugemp]:

$$= (-Wemp/Nlug + Mlugemp/(Rlug \cdot Nlug/2))$$

$$= (-10462/2 + 2814327/(269 \cdot 2/2))$$

$$= 5221.08 \text{ N}$$

Bottom Support Plate Bending Stress (Point Load) [Spl]:

$$= (Flugemp/2 \cdot Dgp/2) / ((Wpl/6) \cdot Tpl^2) \text{ (Uplift Condition)}$$

$$= (5221/2 \cdot 126/2) / ((170/6) \cdot 18^2)$$



= 17.92 MPa

Bottom Plate Required Thickness (Point Load) 5.9862 mm

Bottom Plate required thickness for Uplift per ADM S 3/4 [trUp]:

$$= 0.72 \cdot \sqrt{(\text{Flugemp} \cdot \text{Dgp} / ((\text{Wpl} - \text{dt}) \cdot \text{Spa}))}$$

$$= 0.72 \cdot \sqrt{(5221 \cdot 126 / ((170 - 27.2) \cdot 162))}$$

$$= 3.84 \text{ mm}$$

Bearing Area [Ba]:

$$= \text{Lpl} \cdot \text{Wfb}$$

$$= 150 \cdot 204$$

$$= 30600.00 \text{ mm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot \text{Flug}/\text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 0.21 \cdot 18445/30600 \cdot 204^2/18^2$$

$$= 16.53 \text{ MPa}$$

Bottom Plate Required Thickness (Uniform Load) 5.7505 mm

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot \text{Dgp} \cdot \sqrt{(\text{Flug} / (\text{Lpl} \cdot \text{Wfb})/\text{Spa})}$$

$$= 0.71 \cdot 126 \cdot \sqrt{(18445/(150 \cdot 204)/162)}$$

$$= 5.457 \text{ mm}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot \text{Ylug}$$

$$= 2/3 \cdot 243$$

$$= 162.00 \text{ MPa}$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (\text{Flug}/2) / (\text{Wgp} \cdot \text{Tgp})$$

$$= (18445/2) / (128 \cdot 12)$$

$$= 6.00 \text{ MPa}$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 2.9346 mm

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (50)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (50) / (8 \cdot 127) - (50^3) / (8 \cdot 127^3))$$

$$= 124.13 \text{ MPa}$$

Dimension [Gtp]:

$$= 2 \cdot \text{Wgp} - \text{Wpl}$$

$$= 2 \cdot 128 - 170$$

$$= 86.0 \text{ mm}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((\text{Wpl} - \text{Gtp}) / \text{Hgp})$$

$$= \pi / 2 - \text{atan}((170 - 86) / 150)$$

$$= 60.8 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= \text{Flug} (3 \cdot \text{Dlug} - \text{Wpl}) / (\text{Tgp} \cdot \text{Wpl}^2 \cdot (\text{sin}(\alpha G))^2)$$

$$= 18445 (3 \cdot 185 - 170) / (12 \cdot 170^2 \cdot (\text{sin}(60.8))^2)$$

$$= 26.90 \text{ MPa}$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1/18000) \cdot (\text{Hgp} / \text{sin}(\alpha G) / (0.289 \cdot \text{Tgp}))^2)$$

$$= 18000 / (1 + (1/18000) \cdot (150 / \text{sin}(60.8) / (0.289 \cdot 12))^2)$$

$$= 109.20 \text{ MPa}$$



Determine the required bolt area due to Uplift [Abr]:

= Flug,ten / Sabolt

= 6103/172

= 35.48 mm²

The Area available in a single bolt : 312.748 mm² [OK]

Review notes about nozzle loadings and supports in the Warnings report.

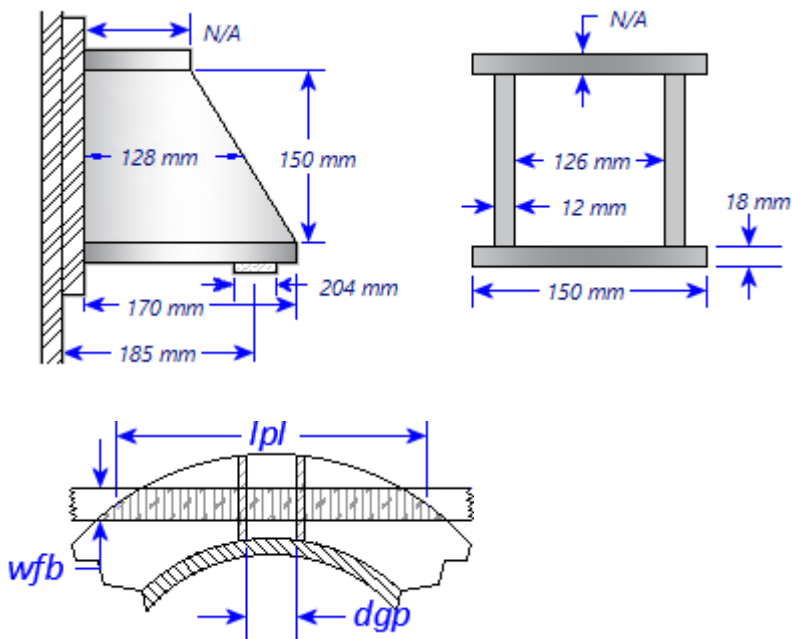
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Support Lug Calculations: Test Case

Input Data: Support Lugs

Type of Geometry : Gussets with No Top Plate
 Number of Support Lugs Nlug 2
 Distance from Vessel OD to Lug Contact Point Dlug 185.0000 mm
 Lug Support Force Bearing Width Wfb 204.0000 mm
 Lug Material SA-516 70 +IT
 Lug Yield Stress 260.00 MPa
 Radial Width of bottom Support Lug Plate Wpl 170.0000 mm
 Effective Force Bearing Length Lpl 150.0000 mm
 Thickness of bottom Support Lug Plate Tpl 18.0000 mm
 Distance between Gussets Dgp 126.0000 mm
 Mean Width of Gusset Plate Wgp 128.0000 mm
 Height of Gusset Plate Hgp 150.0000 mm
 Thickness of Gusset Plate Tgp 12.0000 mm

Bolt Material SA-193 B7
 Bolt Allowable Stress at Design Temperature 172.00 MPa
 Thread Series TEMA Metric
 Bolt Diameter 24.00 mm



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug 1419226.6 N-mm
 Weight Load at the top of one Lug 6445.9 N

Force on one Lug, Test Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$

$$= (12892/2 + 1419226/(269 \cdot 2/2))$$

$$= 11716.80 \text{ N}$$

Force on one lug in the Empty Condition [Flugemp]:

$$= (-Wemp/Nlug + Mlugemp/(Rlug \cdot Nlug/2))$$

$$= (-10462/2 + 2814327/(269 \cdot 2/2))$$

$$= 5221.08 \text{ N}$$

Bottom Support Plate Bending Stress (Point Load) [Spl]:

$$= (Flugemp/2 \cdot Dgp/2) / ((Wpl/6) \cdot Tpl^2) \text{ (Uplift Condition)}$$

$$= (5221/2 \cdot 126/2) / ((170/6) \cdot 18^2)$$



= 17.92 MPa

Bottom Plate Required Thickness (Point Load) 5.7872 mm

Bottom Plate required thickness for Uplift per ADM S 3/4 [trUp]:

$$= 0.72 \cdot \sqrt{(\text{Flugemp} \cdot \text{Dgp} / ((\text{Wpl} - \text{dt}) \cdot \text{Spa}))}$$

$$= 0.72 \cdot \sqrt{(5221 \cdot 126 / ((170 - 27.2) \cdot 173))}$$

$$= 3.71 \text{ mm}$$

Bearing Area [Ba]:

$$= \text{Lpl} \cdot \text{Wfb}$$

$$= 150 \cdot 204$$

$$= 30600.00 \text{ mm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot \text{Flug} / \text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 0.21 \cdot 11717 / 30600 \cdot 204^2 / 18^2$$

$$= 10.50 \text{ MPa}$$

Bottom Plate Required Thickness (Uniform Load) 4.4309 mm

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot \text{Dgp} \cdot \sqrt{(\text{Flug} / (\text{Lpl} \cdot \text{Wfb}) / \text{Spa})}$$

$$= 0.71 \cdot 126 \cdot \sqrt{(11717 / (150 \cdot 204) / 173)}$$

$$= 4.205 \text{ mm}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot \text{Ylug}$$

$$= 2/3 \cdot 260$$

$$= 173.33 \text{ MPa}$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (\text{Flug} / 2) / (\text{Wgp} \cdot \text{Tgp})$$

$$= (11717 / 2) / (128 \cdot 12)$$

$$= 3.81 \text{ MPa}$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 2.5340 mm

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (50)^2 / (2 \cdot 124^2)) 260 / (5/3 + 3 \cdot (50) / (8 \cdot 124) - (50^3) / (8 \cdot 124^3))$$

$$= 132.00 \text{ MPa}$$

Dimension [Gtp]:

$$= 2 \cdot \text{Wgp} - \text{Wpl}$$

$$= 2 \cdot 128 - 170$$

$$= 86.0 \text{ mm}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((\text{Wpl} - \text{Gtp}) / \text{Hgp})$$

$$= \pi / 2 - \text{atan}((170 - 86) / 150)$$

$$= 60.8 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= \text{Flug} (3 \cdot \text{Dlug} - \text{Wpl}) / (\text{Tgp} \cdot \text{Wpl}^2 \cdot (\text{sin}(\alpha G))^2)$$

$$= 11717 (3 \cdot 185 - 170) / (12 \cdot 170^2 \cdot (\text{sin}(60.8))^2)$$

$$= 17.09 \text{ MPa}$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1/18000) \cdot (\text{Hgp} / \text{sin}(\alpha G) / (0.289 \cdot \text{Tgp}))^2)$$

$$= 18000 / (1 + (1/18000) \cdot (150 / \text{sin}(60.8) / (0.289 \cdot 12))^2)$$

$$= 109.20 \text{ MPa}$$



Determine the required bolt area due to Uplift [Abr]:

= Flugemp / Sabolt
= 5221/172
= 30.36 mm²

The Area available in a single bolt : 312.748 mm² [OK]

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N4 1"

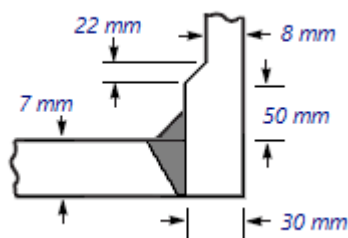
From: 10

Pressure for Reinforcement Calculations	P	5.7204	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Elliptical Head	D	154.30	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	7.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-350 LF2	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	204.7870	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N4 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 25.400 mm.
 Actual Thickness Used in Calculation 8.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 5.72 \cdot 160 \cdot 0.87 / (2 \cdot 118 \cdot 1 - 0.2 \cdot 5.72)$
 $= 3.3935$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.72 \cdot 15.7 / (138 \cdot 1 - 0.6 \cdot 5.72)$
 $= 0.6674$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 15.7$, less than the hub height, use (e-2)
 $= 39.2500$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	70.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6674 mm
Wall Thickness per UG16(b),	tr16b	= 4.5875 mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.7201 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.7201 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875 mm
Wall Thickness per table UG-45	tb3	= 6.1200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.12, \max(6.72, 4.59)]$
 $= 6.1200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.67, 6.12)$
 $= 6.1200$ mm

Available Nozzle Neck Thickness = 8.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	82.5,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.5 MPa	Passed
Occasional	:	7.7,	Allowable	:	183.5 MPa	Passed
Shear	:	58.3,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Weld Size Calculations, Description: N4 1"

Intermediate Calc. for nozzle/shell Welds Tmin 5.2353 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.6647 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.4536 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 215.2406 mm

Input Echo, WRC107/537 Item 1, Description: N4 1" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	3.0000 mm
Vessel Diameter	Dv	277.740 mm
Vessel Thickness	Tv	7.000 mm
Design Temperature	T1	80.0 C
Vessel Material		SA-333 6
Vessel UNS Number		K03006
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	3.0000 mm
Nozzle Diameter	Dn	25.400 mm
Nozzle Thickness	Tn	30.000 mm
Nozzle Material		SA-350 LF2
Nozzle UNS Number		K03011
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa
Design Internal Pressure	Dp	5.720 MPa
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

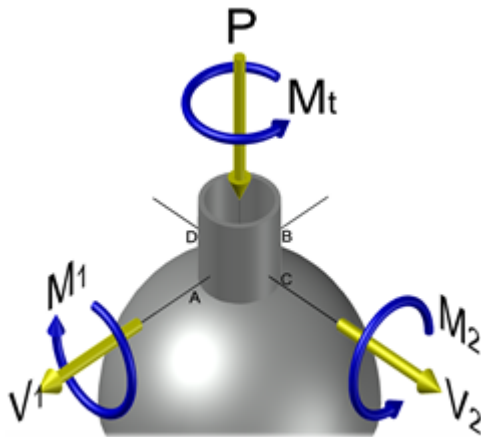
Radial Load (SUS)	P	-1500.0 N
Longitudinal Shear (SUS)	(V1) V1	1300.0 N
Circumferential Shear (SUS)	(Vc) V2	1000.0 N
Circumferential Moment (SUS)	(Mc) M1	200000.0 N-mm

Longitudinal Moment (SUS) (M1) M2 300000.0 N-mm
 Torsional Moment (SUS) Mt 400000.0 N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 85.4 + 2 \cdot 1.65 \cdot \sqrt{(143.87 (7.0 - 3.0))}$
 $= 164.564 \text{ mm}$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-1500.0	N
Circumferential Shear	(VC)	V2	1000.0	N
Longitudinal Shear	(VL)	V1	1300.0	N
Circumferential Moment	(MC)	M1	200000.0	N-mm
Longitudinal Moment	(ML)	M2	300000.0	N-mm
Torsional Moment		MT	400000.0	N-mm

Unitless Prm: U = 1.78 TAU = 5.00 (1.08) RHO = 0.25 (0.15)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02040	(A,B,C,D)
$M(x) / P$	SP 1	0.01959	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.01844	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02291	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.01844	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02291	(A,B,C,D)
$N(y) * T / P$	SP 1	0.02925	(A,B,C,D)
$M(y) / P$	SP 1	0.00560	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02330	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.00695	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02330	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.00695	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Rad. Bend. P		11.0	-11.0	11.0	-11.0	11.0	-11.0	11.0	-11.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-9.6	-9.6	9.6	9.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-71.6	71.6	71.6	-71.6
Rad. Memb. ML		-14.4	-14.4	14.4	14.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-107.4	107.4	107.4	-107.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-108.9	83.9	134.7	-102.1	-68.3	52.9	94.1	-71.1
Tang. Memb. P		2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Tang. Bend. P		3.1	-3.1	3.1	-3.1	3.1	-3.1	3.1	-3.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-12.1	-12.1	12.1	12.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-21.7	21.7	21.7	-21.7
Tang. Memb. ML		-18.2	-18.2	18.2	18.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-32.6	32.6	32.6	-32.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-44.9	13.9	56.7	-14.8	-28.0	9.2	39.7	-10.0
Shear VC		1.9	1.9	-1.9	-1.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.4	-2.4	2.4	2.4
Shear MT		8.7	8.7	8.7	8.7	8.7	8.7	8.7	8.7
Tot. Shear		10.6	10.6	6.9	6.9	6.3	6.3	11.1	11.1
Str. Int.		110.6	85.4	135.3	102.6	69.2	53.8	96.3	73.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		101.5	101.5	101.5	101.5	101.5	101.5	101.5	101.5
Rad. Pl (SUS)		-12.5	-12.5	16.3	16.3	-7.7	-7.7	11.5	11.5
Rad. Q (SUS)		-96.4	96.4	118.4	-118.4	-60.6	60.6	82.6	-82.6
Long. Pm (SUS)		101.5	101.5	101.5	101.5	101.5	101.5	101.5	101.5
Long. Pl (SUS)		-15.5	-15.5	21.0	21.0	-9.4	-9.4	14.9	14.9
Long. Q (SUS)		-29.4	29.4	35.7	-35.7	-18.6	18.6	24.9	-24.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.9	1.9	-1.9	-1.9	-2.4	-2.4	2.4	2.4
Shear Q (SUS)		8.7	8.7	8.7	8.7	8.7	8.7	8.7	8.7
Pm (SUS)		101.5	101.5	101.5	101.5	101.5	101.5	101.5	101.5
Pm+Pl (SUS)		89.9	89.9	123.1	123.1	95.5	95.5	117.6	117.6
Pm+Pl+Q (Total)		67.4	186.9	236.8	88.4	74.5	155.3	197.8	93.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	101.50	118.00	Passed
Pm+Pl (SUS)	123.11	177.00	Passed
Pm+Pl+Q (TOTAL)	236.82	353.99	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: N1 2"

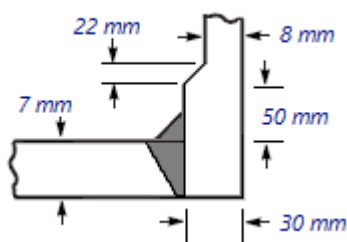
From: 40

Pressure for Reinforcement Calculations	P	5.7098	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1047.23	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-350 LF2	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 50.800 mm.
 Actual Thickness Used in Calculation 8.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9944$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 5.71)$
 $= 1.2050$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22 / 0.5773$
 $= 38.1051$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N1 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.2050	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9944	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9944	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 7.5200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.52, \max(6.99, 4.59)]$
 $= 6.9944$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.2, 6.99)$
 $= 6.9944$ mm

Available Nozzle Neck Thickness = 8.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	39.2,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	305.8 MPa	Passed
Occasional	:	14.9,	Allowable	:	183.5 MPa	Passed
Shear	:	19.5,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of

the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Weld Size Calculations, Description: N1 2"

Intermediate Calc. for nozzle/shell Welds Tmin 4.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 23.4568 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.3068 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 51.42 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm

Input Echo, WRC297 Item 1, Description: N1 2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	154.300	mm
Shell Thickness	7.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-333 6	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*



Design Internal Pressure	Dp	5.71	MPa
Radial Load	P	-1500.00	N
Circumferential Shear	Vc	1000.00	N
Longitudinal Shear	Vl	1300.00	N
Circumferential Moment	Mc	200000.02	N-mm
Longitudinal Moment	Ml	300000.00	N-mm
Torsional Moment	Mt	400000.03	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.658) is > 0.5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (3.693) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 110.8 + 2 \cdot 1.65 \cdot \sqrt{(82.15(7.0 - 3.0))}$
 $= 170.620 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	164.300 mm
Nozzle Outside Diameter	(d) =	110.800 mm
Vessel Thickness used	(T) =	4.000 mm
Nozzle Thickness used	(t) =	27.000 mm
T / t	=	0.148
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.322$$

Nr/P	=	0.048
Mr/P	=	0.077
M0/P	=	0.023
N0/P	=	0.041
MrD/Mc	=	0.250
NrDL/Mc	=	0.094
M0d/Mc	=	0.075
N0DL/Mc	=	0.071
MrD/Ml	=	0.060
NrDL/Ml	=	0.049
M0D/Ml	=	0.018
N0DL/Ml	=	0.042

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-7	-7	7	7
Outplane Bending (Ml)	-18	18	18	-18
Normal Pressure Stress	111	117	111	117
Outplane Stress Summary	102	119	154	96

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4



Inplane Bending (P)	43	-43	43	-43
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-61	61	61	-61
Inplane Pressure Stress	55	55	55	55

Inplane Stress Summary	33	70	173	-35
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	6	6	3	3
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	103	120	173	132
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-50	50	50	-50
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	55	55	55	55

Outplane Stress Summary	14	89	131	4
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	43	-43	43	-43
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-169	169	169	-169
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	111	117	111	117

Inplane Stress Summary	-20	237	339	-79
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	7	7
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
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Two * Max Shear Stress	35	237	339	85
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	11	4	11

Hoop Stress Summary	1	8	15	22
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	0	0	6	3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	2	8	15	22
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	11	4	11

Hoop Stress Summary	0	7	16	23
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-1	3	7	0
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	2	7	16	23

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	111	117	111	117	111	117	111	117
Circ. Pl (SUS)	-4	-4	10	10	-6	-6	14	14
Circ. Q (SUS)	-5	5	31	-31	-126	126	212	-212
Long. Pm (SUS)	55	55	55	55	55	55	55	55
Long. Pl (SUS)	-4	-4	12	12	-4	-4	10	10
Long. Q (SUS)	-18	18	104	-104	-37	37	63	-63
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	111.0	117.0	111.0	117.0	111.0	117.0	111.0	117.0
Pm+Pl (SUS)	107.0	113.0	121.0	127.0	105.0	111.0	125.0	131.0
Pm+Pl+Q (Total)	102.5	118.7	171.8	133.2	35.9	237.1	337.2	83.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	118.00	Passed
Pm+Pl (SUS)	131.02	177.00	Passed
Pm+Pl+Q (TOTAL)	337.17	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	11	4	11	4	11	4	11
Circ. Pl (SUS)	-4	-4	10	10	-4	-4	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	0	0	0	0	-3	3	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	4.0	11.0	4.0	11.0	4.0	11.0	4.0	11.0
Pm+Pl (SUS)	0.0	7.0	14.0	21.0	1.0	7.0	14.0	21.0
Pm+Pl+Q (Total)	0.0	7.0	14.0	21.0	2.0	7.0	14.0	21.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	138.00	Passed
Pm+Pl (SUS)	21.00	207.00	Passed
Pm+Pl+Q (TOTAL)	21.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1A 1"

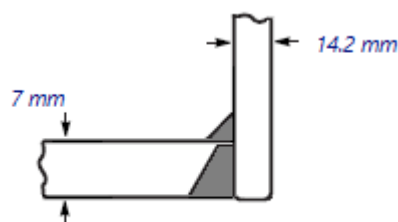
From: 40

Pressure for Reinforcement Calculations	P	5.7068	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1350.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1A 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9922 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.71 / (138 \cdot 1)] - 1))$
 $= 0.6629 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6629	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9922	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9922	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(6.99, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K1A 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm



This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 31.40 mm

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Input, Nozzle Desc: K1B 1"

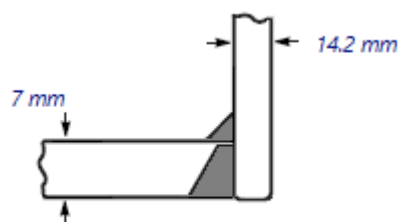
From: 40

Pressure for Reinforcement Calculations	P	5.7161	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1B 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.72 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.72)$
 $= 3.9989 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.72 / (138 \cdot 1)] - 1))$
 $= 0.6640 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1B 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6640	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9989	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9989	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(7, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K1B 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm



This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 31.40 mm

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Input, Nozzle Desc: K2A 1"

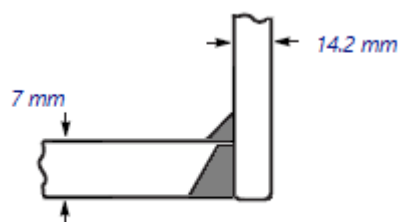
From: 40

Pressure for Reinforcement Calculations	P	5.7068	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1350.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2A 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9922 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.71 / (138 \cdot 1)] - 1))$
 $= 0.6629 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6629	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9922	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9922	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(6.99, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K2A 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 401-T-32001_MDS-Rev 01 Page 140 of 179
Nozzle Calcs.: K2A 1" Nozl: 37 10:01am Jul 20,2026



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Input, Nozzle Desc: K2B 1"

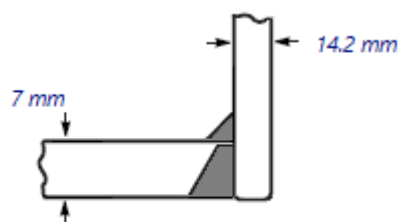
From: 40

Pressure for Reinforcement Calculations	P	5.7161	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2B 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.72 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.72)$
 $= 3.9989 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.72 / (138 \cdot 1)] - 1))$
 $= 0.6640 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2B 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6640	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9989	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9989	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(7, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K2B 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 401-T-32001_MDS-Rev 01 Page 143 of 179
Nozzle Calcs.: K2B 1" Nozl: 38 10:01am Jul 20,2026



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Input, Nozzle Desc: K3A 1"

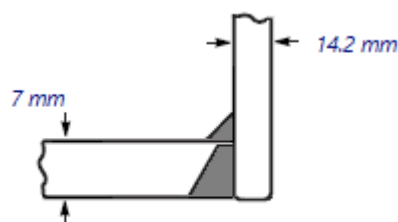
From: 40

Pressure for Reinforcement Calculations	P	5.7068	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1350.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3A 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9922 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.71 / (138 \cdot 1)] - 1))$
 $= 0.6629 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6629	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9922	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9922	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(6.99, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K3A 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 401-T-32001_MDS-Rev 01 Page 146 of 179
Nozzle Calcs.: K3A 1" Nozl: 39 10:01am Jul 20,2026



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Input, Nozzle Desc: K3B 1"

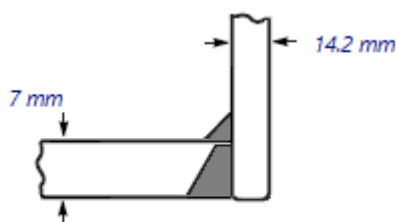
From: 40

Pressure for Reinforcement Calculations	P	5.7161	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3B 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.72 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.72)$
 $= 3.9989 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.72 / (138 \cdot 1)] - 1))$
 $= 0.6640 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3B 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6640	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9989	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9989	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(7, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K3B 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 401-T-32001_MDS-Rev 01 Page 149 of 179
Nozzle Calcs.: K3B 1" Nozl: 40 10:01am Jul 20,2026



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Input, Nozzle Desc: K3C 1"

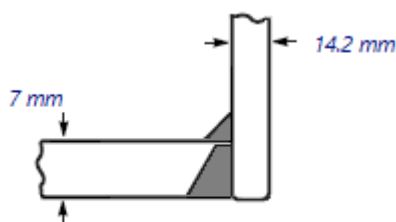
From: 40

Pressure for Reinforcement Calculations	P	5.7068	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1350.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3C 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9922 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.71 / (138 \cdot 1)] - 1))$
 $= 0.6629 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3C 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6629	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9922	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9922	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(6.99, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K3C 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm



This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 23.05 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 31.40 mm

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Input, Nozzle Desc: K3D 1"

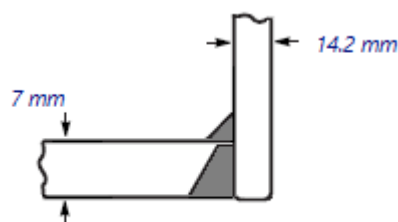
From: 40

Pressure for Reinforcement Calculations	P	5.7161	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3D 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.72 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.72)$
 $= 3.9989 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.72 / (138 \cdot 1)] - 1))$
 $= 0.6640 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3D 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6640	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9989	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9989	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(7, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K3D 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm



This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 23.05 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 31.40 mm

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Input, Nozzle Desc: K3E 1"

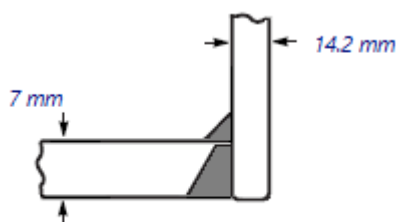
From: 40

Pressure for Reinforcement Calculations	P	5.7068	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1350.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3E 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9922 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.71 / (138 \cdot 1)] - 1))$
 $= 0.6629 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3E 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6629	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9922	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9922	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(6.99, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K3E 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm





Input, Nozzle Desc: K3F 1"

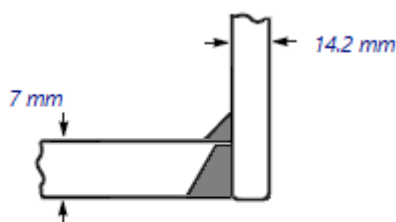
From: 40

Pressure for Reinforcement Calculations	P	5.7161	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3F 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.72 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.72)$
 $= 3.9989 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 15.7 \left(\exp \left(\frac{5.72}{138 \cdot 1} \right) - 1 \right)$
 $= 0.6640 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3F 1".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6640	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9989	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9989	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(7, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -46 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Weld Size Calculations, Description: K3F 1"

Intermediate Calc. for nozzle/shell Welds Tmin 4.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$



Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 4.8505 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm

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Input, Nozzle Desc: K4B 1"

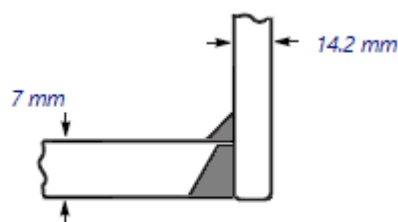
From: 40

Pressure for Reinforcement Calculations	P	5.7064	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1597.23	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4B 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9919 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.71 / (138 \cdot 1)] - 1))$
 $= 0.6628 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4B 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6628	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9919	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9919	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(6.99, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K4B 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 401-T-32001_MDS-Rev 01 Page 164 of 179
Nozzle Calcs.: K4B 1" Nozl: 45 10:01am Jul 20,2026



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Input, Nozzle Desc: N2 1"

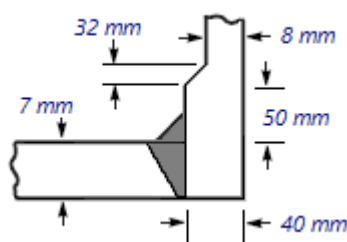
From: 100

Pressure for Reinforcement Calculations	P	5.7064	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4800.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	32.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 25.400 mm.
 Actual Thickness Used in Calculation 8.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9919$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 15.7 / (138 \cdot 1 - 0.6 \cdot 5.71)$
 $= 0.6657$ mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.7 / 0.5773$
 $= 18.5330$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6657	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9919	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9919	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.1200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.12, \max(6.99, 4.59)]$
 $= 6.1200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.67, 6.12)$
 $= 6.1200$ mm

Available Nozzle Neck Thickness = 8.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	82.4,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.6 MPa	Passed
Occasional	:	7.7,	Allowable	:	183.5 MPa	Passed
Shear	:	58.3,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of



the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Weld Size Calculations, Description: N2 1"

Intermediate Calc. for nozzle/shell Welds Tmin 4.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 6.6790 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.6790 mm

Input Echo, WRC297 Item 1, Description: N2 1" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	154.300	mm
Shell Thickness	7.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-333 6	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	25.400	mm
Nozzle Thickness	40.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	5.71	MPa
Radial Load	P	-1500.00	N
Circumferential Shear	Vc	1000.00	N
Longitudinal Shear	Vl	1300.00	N
Circumferential Moment	Mc	200000.02	N-mm
Longitudinal Moment	Ml	300000.00	N-mm
Torsional Moment	Mt	400000.03	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



Warning - The ratio, D_n/D_v (.626) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, D_n/T_n (2.635) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 105.4 + 2 \cdot 1.65 \cdot \sqrt{(82.15(7.0 - 3.0))}$
 $= 165.220 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 164.300 mm
 Nozzle Outside Diameter (d) = 105.400 mm
 Vessel Thickness used (T) = 4.000 mm
 Nozzle Thickness used (t) = 37.000 mm
 $T / t = 0.108$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.111$$

Nr/P = 0.051
 Mr/P = 0.081
 M0/P = 0.024
 N0/P = 0.036
 MrD/Mc = 0.251
 NrDL/Mc = 0.102
 M0d/Mc = 0.075
 N0DL/Mc = 0.062
 MrD/Ml = 0.060
 NrDL/Ml = 0.057
 M0D/Ml = 0.018
 N0DL/Ml = 0.038

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	111	117	111	117
Outplane Stress Summary	101	120	154	94

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-64	64	64	-64
Inplane Pressure Stress	55	55	55	55
Inplane Stress Summary	31	69	180	-39

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	4	4
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	102	121	181	134
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-53	53	53	-53
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	55	55	55	55

Outplane Stress Summary	11	91	133	0
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-178	178	178	-178
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	111	117	111	117

Inplane Stress Summary	-28	243	352	-90
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	7	7
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	41	243	353	91
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	8	1	8

Hoop Stress Summary	-2	4	11	18
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-1	-1	4	2
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	2	6	11	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	8	1	8

Hoop Stress Summary	-2	4	12	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-2	0	4	0
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	2	4	12	18

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	111	117	111	117	111	117	111	117
Circ. Pl (SUS)	-3	-3	9	9	-8	-8	16	16
Circ. Q (SUS)	-6	6	32	-32	-133	133	223	-223
Long. Pm (SUS)	55	55	55	55	55	55	55	55
Long. Pl (SUS)	-6	-6	14	14	-4	-4	10	10
Long. Q (SUS)	-19	19	109	-109	-40	40	66	-66
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	111.0	117.0	111.0	117.0	111.0	117.0	111.0	117.0
Pm+Pl (SUS)	108.0	114.0	120.0	126.0	103.0	109.0	127.0	133.0
Pm+Pl+Q (Total)	102.5	120.7	178.6	134.2	41.8	242.1	350.2	90.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	118.00	Passed
Pm+Pl (SUS)	133.01	177.00	Passed
Pm+Pl+Q (TOTAL)	350.16	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	8	1	8	1	8	1	8
Circ. Pl (SUS)	-3	-3	9	9	-4	-4	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	0	0	0	0	-1	1	1	-1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	8.0	1.0	8.0	1.0	8.0	1.0	8.0
Pm+Pl (SUS)	2.0	7.0	10.0	17.0	3.0	5.0	11.0	18.0
Pm+Pl+Q (Total)	2.0	7.0	10.0	17.0	3.0	4.0	11.0	18.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	18.00	207.00	Passed
Pm+Pl+Q (TOTAL)	18.00	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K4A 1"

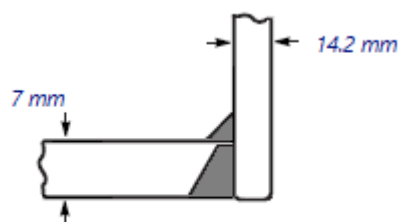
From: 100

Pressure for Reinforcement Calculations	P	5.7064	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-333 6	+IT
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	154.30	mm
Shell Finished (Minimum) Thickness	t	7.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4800.00	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		25.4000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	252.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4A 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	25.400	mm.
Actual Thickness Used in Calculation	14.224	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 80.2 / (118 \cdot 1 - 0.6 \cdot 5.71)$
 $= 3.9919 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([5.71 / (138 \cdot 1)] - 1))$
 $= 0.6628 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	62.8000	mm
Parallel to Vessel Wall, opening length	d	31.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.6628	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 6.9919	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 6.9919	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(6.99, 4.59)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.66, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Weld Size Calculations, Description: K4A 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	4.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.8000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 4.8505 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.8505 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 401-T-32001_MDS-Rev 01 Page 175 of 179
Nozzle Calcs.: K4A 1" Nozl: 47 10:01am Jul 20,2026



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Input, Nozzle Desc: N3 2"

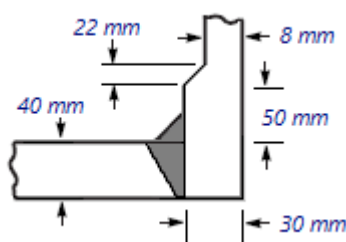
From: 120

Pressure for Reinforcement Calculations	P	5.7064	MPa
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-350 LF2 +IT	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Outside Diameter of Bolted Blind Flange	D	317.50	mm
Head Finished (Minimum) Thickness	t	40.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		40.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.000 mm.
 Actual Thickness Used in Calculation 8.000 mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.71 \cdot 28 / (138 \cdot 1 - 0.6 \cdot 5.71)$
 $= 1.1873$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22 / 0.5773$
 $= 38.1051$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	140.0000	mm
Parallel to Vessel Wall	Rn+tn+t	70.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.6051	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3 2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.1873	mm
Wall Thickness per UG16(b),	tr16b	= 4.5875	mm
Wall Thickness, shell/head, internal pressure	trb1	= 38.7459	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 38.7459	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5875	mm
Wall Thickness per table UG-45	tb3	= 7.5200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.52, \max(38.7, 4.59)]$
 $= 7.5200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.19, 7.52)$
 $= 7.5200$ mm

Available Nozzle Neck Thickness = 8.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :



Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = $5.71/15.31 = 0.373$

Weld Size Calculations, Description: N3 2"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0500 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.3500 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 190.0508 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

Please review all reports carefully!

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